

Capability Scaling-Down Laws of LLM Compression in Agent Systems

Anonymous authors

Paper under double-blind review

Abstract

We systematically investigate whether information observable before compression can predict capability-conditioned loss after pruning, quantization, and distillation. The endpoint is a per-capability conditional loss (math, code, multi-hop QA) measured relative to the dense source or, for distillation, the initial student; the inputs are the source size, its pretraining tokens, its dense capability losses, and the intervention setting. We separate transfer across source-model states from prediction at unseen intervention settings, evaluate each candidate against baselines with the same information budget, and freeze predictions before new sources are measured. On a heterogeneous panel of 12 public models from five series, responses are strongly capability- and series-specific: QA is the least-damaged capability in Qwen3 and OLMo-3-32B and the most-damaged one in Gemma at mild pruning, and 3-bit quantization collapses 12 of 14 models but not OLMo-3-32B. On a controlled Pythia panel, the dense capability vector carries the prediction; source-conditioned pruning predictors beat source-free curves at seen densities and at interpolated densities on the development sources, match them at a new in-range size, and fail at density 0.55 and at a source five times larger; a shared power form adds nothing over a per-density regression. Quantization gains appear only in the collapse regime and depend on the fitting protocol, whereas adding the quantization group size as a second axis lets a low-order relation predict unseen bit-widths and group sizes on seen states; for distillation a reuse-count coordinate helps QA on an unseen pool with a budget-dependent bias, and a constant is best for math and code. The results identify conditional predictive relationships and their tested limits, providing an empirical basis for developing capability-conditioned scaling-down laws.

1 Introduction

Deployed language-model agents are specialists: a coding agent depends on code synthesis and tool use, a retrieval assistant on multi-hop question answering, and both pay for every token of inference. The practical question is which model covers a workload’s capability profile at the lowest cost, and the usual answer starts from a strong dense source and scales it down by pruning, quantization, or distillation. Compression does not shrink capabilities uniformly: at the same size or bit-width, different methods and source families preserve different capabilities, improve some while collapsing others, and hit cliffs at different strengths (§5.1). Aggregate perplexity hides this, so choosing a route today means running each candidate configuration through capability benchmarks.

This paper asks whether that search can be predicted from what is observable before compression. We fix a per-capability conditional loss as the endpoint and write every prediction relative to a reference measured beforehand, the dense source for pruning and quantization and the initial student for distillation (§3). The admitted inputs are the source size, its pre-training token count where disclosed, its dense capability losses, and the intervention setting. The question is then concrete: can pre-compression model-state information and interven-

tion settings predict capability-conditioned loss, and over which source and intervention ranges do those predictions transfer?

A predictor can fail because the source information is useless, because the functional form is wrong, or because the test lies outside the range in which anything was fit, and a single accuracy number blurs the three. We therefore test two axes separately, transfer across source states at a fixed intervention and prediction at unseen intervention settings, and compare every candidate with baselines that use the same information: a source-free curve or per-configuration constant asks whether the source state helps at all, a no- D_0 variant asks what training tokens add, and same-input alternatives ask whether a particular form adds anything (§4). New-source tests are frozen prospectives, committed before the compressed model is measured, and a failed prospective keeps its label.

The evidence has three layers (§5). On a heterogeneous panel of 12 public models from five series, the response to compression is capability- and series-specific: QA is the least-damaged capability in Qwen3 and OLMo-3-32B, where it improves under moderate pruning, and the most-damaged one in the Gemma series at mild densities; 3-bit quantization collapses 12 of 14 models but not OLMo-3-32B. On a controlled Pythia panel, the dense capability vector carries the prediction and training tokens add little beyond it; source-conditioned pruning predictors beat source-free curves at seen densities and at interpolated densities on the development sources, mostly for math and code, match them at a new in-range size, and fail at density 0.55 and at a source five times larger, in both arms; a shared power form adds nothing over a per-density regression. Quantization gains appear only in the collapse regime and depend on the fitting protocol, but with the quantization group size as a second axis a low-order relation predicts unseen bit-widths and group sizes on seen states. For distillation, a reuse-count coordinate helps QA on an unseen pool with a budget-dependent bias, and a constant is best for math and code. Method-agnostic laws in storage or active-parameter ratios fail a matched-storage test.

Contributions. (1) A capability-conditioned prediction problem with an explicit information budget, one loss interface for three compression methods, and a two-axis validation design (§3, §4). (2) Candidate relations per arm, compared on the same inputs with same-budget baselines and frozen prospective tests on new sources, with their measured range of validity (§5.2–5.4). (3) An empirical account of when transferable prediction is possible and where it fails, with reproducible measurements, registers, and evaluation code (§5.5). Dense-reference scaling laws are prior work we use, the Fisher implementation is infrastructure, and method selection is discussed but not validated.

2 Related Work

Compression scaling laws with a loss endpoint. Most compression laws predict an aggregate loss: sparsity as a multiplicative factor on the size term (Frantar et al., 2024), precision through an effective-parameter count with a degradation term that grows with pretraining tokens (Kumar et al., 2025), formats used during training through one capacity scalar (Panferov et al., 2025), and a distilled student through a broken power law in teacher loss (Busbridge et al., 2025). Closest to our pruning arm, the P^2 law (Chen et al., 2025) predicts the post-training loss of a pruned model from its size and loss before pruning, the pruning rate, and the post-training tokens, validated on held-out tokens, sizes, and rates; source-state conditioning is thus not new in itself. None of these fit or validate a form on per-capability measurements (Busbridge et al. (2025) report downstream accuracy only as a proxy check).

Laws with downstream endpoints. Sengupta et al. (2025) fit pruning laws on test loss and averaged zero-shot accuracy against the compression ratio and recovery tokens; the revised version of the same work adds rolling hold-out over pruning ratios, zero- and one-shot transfer to unseen models, and per-task exponents. Its inputs are the unpruned score and the pruning ratio, with no source-state or training-history term, and its endpoint is an accuracy or an inverse log-perplexity. Bit-width laws for zero-shot accuracy exist at large scale (Dettmers & Zettlemoyer, 2023), and Zhou et al. (2026) stratify post-training

quantization laws into memorization, application, and reasoning strata for one quantizer, validated on a held-out larger model and a second family; those laws carry no source-state input and no recovery axis. Task-specific distillation has been studied as curves over dataset size, compression ratio, and supervision format without a fitted law (Ghita et al., 2026). Relative to this line, the present study keeps a per-capability conditional loss as the endpoint for three methods at once, admits the dense capability vector and the training-token count as inputs on a panel that separates size from training stage, and reports frozen tests against baselines with the same information, including the tests that fail.

Predictability of downstream capabilities. Downstream accuracy appears predictable mainly through continuous log-likelihood measurements: emergent jumps are largely metric artifacts (Schaeffer et al., 2023), accuracy-space prediction degrades through information-destroying transforms (Schaeffer et al., 2025), and successful pipelines fit loss-space laws with sigmoid or exponential links (Gadre et al., 2025; Bhagia et al., 2025; Owen, 2024; Ruan et al., 2024); even then, linear loss-to-downstream relations hold in a minority of settings (Lourie et al., 2025). The measurement layer of §3.3 follows these lessons.

Capability-selective effects of compression. Pruning disproportionately harms knowledge-intensive tasks (Jaiswal et al., 2024), low-bit quantization disproportionately harms reasoning (Liu et al., 2025), small-magnitude weights encode hard-task knowledge whose removal is not recovered by further training (Yin et al., 2024), and selective low-rank truncation can improve specific behaviors (Sharma et al., 2024). These observations motivate a capability-vector endpoint; the present paper adds per-model, per-configuration measurements across families and a prediction test of the resulting heterogeneity.

Gradients, curvature, and law forms. Second-order weight importance underlies classical and modern pruning (LeCun et al., 1989; Hassibi & Stork, 1992; Frantar & Alistarh, 2023; Sun et al., 2024), Fisher-aligned subspaces improve compression choices (Shihab et al., 2026), and gradient structure localizes skills (Gur-Ari et al., 2018; Zhao et al., 2024; Michaud et al., 2023; Ilharco et al., 2023); we use capability-conditional gradient and Fisher statistics only as mechanism evidence (Appendix B), never as inputs. Surveys document the fragility of purely fitted scaling-law forms (Li et al., 2025; Besiroglu et al., 2024); the forms tested here are low-dimensional candidates compared by held-out error and kept only where they predict.

3 Problem and Measurement

3.1 The prediction problem

We fix a capability set $\mathcal{C}$ (mathematical reasoning, code generation, multi-hop question answering) and one observable per capability, the conditional loss L_c of Eq. equation 2. For a compression method m the target is the post-compression capability vector written relative to a reference measured before the intervention,

$$\hat{L}_{m,c} = L_{c,\text{ref}(m)}^{\text{obs}} + \hat{\Delta}_{m,c}(\mathbf{x}, \mathbf{z}, \boldsymbol{\theta}_m). \quad (1)$$

For pruning and quantization the reference is the dense source, so $\hat{\Delta}_{m,c}$ predicts the damage $\Delta L_c = L_c(\text{compressed}) - L_c(\text{dense})$. For distillation it is the initial student, so the target is the signed transfer $\delta_c = L_c(S_{\text{KD}}) - L_c(S_0)$, and the size gap between teacher and student never enters. The arms share the endpoint and the evaluation protocol and nothing else by assumption; whether one form covers several arms is a question for the data (§5.5).

The inputs fall into three groups. The state $\mathbf{x}$ is what is observable about the source or student beforehand: N_0 , counted throughout as the number of entries of the transformer weight matrices (attention and MLP projections; embeddings, output head, biases, and norms excluded); the pretraining token count D_0 where disclosed; and the measured dense capability vector $\mathbf{L}_0 = (L_{0,\text{math}}, L_{0,\text{code}}, L_{0,\text{qa}})$. Family identity and per-model fitted amplitudes are not basic parameters, and models that use them are said to. The shared descriptors $\mathbf{z}$ are the coordinates on which arms can be compared: retained density, bit-width, and the processed

Budget	Information available	Cost to disclose
K0, metadata only	N_0 , public D_0 , architecture, intervention configuration	metadata availability; D_0 may be undisclosed
K0 + dense anchor	the above plus the dense capability losses $\mathbf{L}_0$, measured once before compression	probe count, one forward pass per probe, native loss units
K1	the above plus one pre-specified measurement of the target at one compression configuration	the calibration configuration and its compression cost
Oracle diagnostic	post-hoc rescaling on the target outcomes	none; diagnoses form misfit, never a prediction

Table 1: Information budgets. Unknown D_0 stays unknown; handling of missing inputs is fixed on the development panel and never inferred from the target response. Predicting $\mathbf{L}_0$ from (N_0, D_0) is the task of dense scaling laws and is not attempted here.

and unique training tokens of a distillation run. The method controls θ_m are the per-arm settings held fixed by protocol (pruning criterion, quantizer, student size, supervision recipe). No shared variable is invented for an arm that lacks one.

3.2 Information budgets

Every predictor is labeled by the information it may use (Table 1). All headline predictions use the second budget; the first is the reference for what training tokens add.

3.3 Operational loss definition

Each capability has a probe set $\mathcal{P}_c$ of prompt and reference pairs drawn once with a fixed seed from held-out benchmark splits (MATH-500, MBPP, and 2WikiMultihopQA as primary benchmarks). The observable is the token-weighted conditional cross-entropy of the reference completion given the prompt,

$$L_c(w) = \frac{1}{T_c} \sum_{(x,y) \in \mathcal{P}_c} \sum_{t=1}^{|y|} -\log p_w(y_t \mid x, y_{<t}), \quad T_c = \sum_{(x,y) \in \mathcal{P}_c} |y|, \quad (2)$$

with prompt tokens masked, sequences truncated to 1024 tokens, the model’s own tokenizer, and the same 64 measurement probes per capability for every state compared within a model. We report native nats per token. Within-model differences ΔL_c on identical token sequences are the only quantities that enter predictors; absolute levels across tokenizers are never placed on one scale.

Three boundaries apply. The observable is the negative log-likelihood of a fixed reference, not a task-success measure: a QA loss can fall while exact-match accuracy does not rise (Appendix A.1). The reference is the full solution; direct-answer and given-reasoning variants differ from it for math and QA (Appendix A.1). Capability specificity is a tested property: the response on an independent secondary benchmark is predicted better from the same capability’s primary response than from another capability’s for math and QA, whereas for code the within-capability gain was not detected at the available resolution (Appendix A.1, Fig. 3). Gradient and Fisher statistics of the dense model appear only as mechanism evidence in the appendix; they are never inputs.

4 Study Design and Predictive Models

4.1 Panels

Three kinds of panel serve three purposes (Appendix Table 4). The heterogeneous panel of 12 finished public models from 5 series (Qwen3, Gemma-3, Gemma-4, Muse, OLMo-3; 270M to 32B) establishes the phenomena and tests how far a relation carries across families whose training history, data, and architecture differ at once; family and provenance are

confounded by construction. The controlled Pythia panel, sizes 160M, 410M, and 1.4B at pretraining steps 16k, 64k, and 143k (nine states), varies the source state along known axes with architecture and data fixed and is the development set of every predictor; new sources for frozen tests are 160M, 410M, and 1.4B at step 96k, 1B at 32k, 96k, and 112k, and 6.9B at 32k and 112k. The distillation panels use the nine Pythia states as students with a fixed teacher pool, and Gemma-3-1B with pools of 75, 225, and 600 traces per domain.

4.2 Predictors that were tested

Every predictor was fit on the development set only, with standardization constants, coefficients, and rules committed to a register before any test measurement (Appendix A.4). Parameter counts are per capability.

Configuration-indicator source regression (source axis). At each intervention configuration q , a density or bit-width, a separate linear model $\widehat{\Delta L}_c(q) = \beta_{c,q} \cdot (1, \log \tilde{N}_0, \tilde{L}_{0,c}, \log \tilde{D}_0)$ in standardized inputs, four coefficients per configuration. It predicts a new source state at a seen configuration and is not a law over the intervention axis; its same-input comparator drops $\log \tilde{D}_0$. It is the predictor frozen for the step-96k tests (register v38) and, per bit-width, the quantization predictor throughout.

Shared power form (pruning, configuration axis). $\widehat{\Delta L}_c(\mathbf{x}, d) = A_c(\mathbf{x}) ((1 - d)/0.3)^{\gamma_c}$ with $A_c(\mathbf{x}) = \beta_c \cdot (1, \log \tilde{N}_0, \tilde{L}_{0,c}, \log \tilde{D}_0)$, five parameters, fit on the nine development states at $d \in \{0.9, 0.8, 0.7, 0.6\}$ (registers v40, v49). Three same-input alternatives use the same inputs with a different strength structure: A1, the power form with $\gamma_c = 1$; A2, the configuration-indicator regression at the four development densities plus a fixed rule (linear interpolation between adjacent predicted anchors, linear extension beyond the boundary pair); and a continuous two-term polynomial in $1 - d$. Three source-free baselines ask whether the source state is worth reading at all: a strength-only curve $A((1 - d)/0.3)^\gamma$ fit on all development rows, the median development curve, and zero change.

Quantization baselines. Per bit-width, the mean and median of the development responses and zero change; the 5-bit prediction of every candidate is a fixed rule, the average of its committed 4- and 6-bit predictions.

Distillation predictors. On the source axis, a linear model in $(\log \tilde{N}_0, \tilde{L}_{0,c}, \log \tilde{D}_0)$, its no- $\tilde{D}_0$ variant, and per-capability constants. On the pool axis, four forms fit on the 18 endpoint checkpoints of pools U75 and U600, a constant, $a + b \log T$, $a + b \log E$, and $a + b \log T + c \log E$ with $E = T/D_U$ the reuse count, all frozen and applied to the six U225 runs (register v41).

4.3 Validation protocol and labels

On the source axis the intervention setting is seen and the source state is not, through leave-one-stage-out folds or a frozen prospective on a new source; on the configuration axis the setting is unseen (densities, bit-widths, pools), with or without a new source. New-source tests follow a fixed order: forms and standardization are committed; the dense capability vector of the new source is measured, since it is a permitted input; every candidate's predictions are written to a versioned file and committed with the hash recorded; only then is the outcome measured. The last round uses two fitting protocols, A on all nine development states and B on states at step 64k or below, so that a 112k test stage is a stage extrapolation under B.

Each comparison carries a four-part origin code (Table 3): whether the candidate prediction was frozen or is a leave-one-out estimate, whether the same-input and the simple baseline were pre-specified with it or computed afterwards, and whether all pre-specified forms are reported. A frozen prediction keeps its label whatever the outcome; where several baselines exist, the strongest per capability is shown and marked as a retrospective choice. The only post-test selection is the per-capability reading of the four distillation forms, labeled as such. Which baseline answers which question: zero change and the constant ask whether the

Model	ΔL_c at $d = 0.7$			math@0.5	d_{math}^*	QA least worst	min QA (d)
	math	code	QA				
Qwen3-0.6B	+0.49	+0.56	-0.26	+4.72	0.60	3/3 0/3	-0.26 (0.70)
Qwen3-1.7B	+0.24	+0.19	-0.51	+3.60	0.60	6/6 0/6	-0.90 (0.62)
Qwen3-4B	-0.00	-0.09	-0.79	+1.31	0.50	5/5 0/5	-1.96 (0.55)
Qwen3-8B [†]	+0.03	+0.10	-0.41	–	none by 0.60	4/4 0/4	-1.02 (0.60)
Qwen3-14B [†]	+0.06	+0.03	-0.45	–	none by 0.60	3/4 0/4	-0.80 (0.60)
Gemma-3-270M	+3.49	+4.04	+2.54	+11.21	0.70	0/2 1/2	+0.21 (0.90)
Gemma-3-1B	+1.31	+1.23	+0.83	+13.78	0.70	0/5 4/5	+0.19 (0.90)
Gemma-3-4B	+0.83	+1.47	+0.24	+7.00	0.60	1/3 2/3	+0.05 (0.90)
Gemma-3-12B	+0.68	+0.46	+0.54	+17.10	0.60	0/3 2/3	+0.07 (0.90)
Gemma-3-27B	+4.04	+4.69	+3.59	+15.21	0.70	1/2 1/2	+0.01 (0.80)
Gemma-4-31B	+0.68	+0.77	+2.89	+16.06	0.60	0/3 3/3	+0.22 (0.90)
Muse-30B	+0.17	+0.14	+0.20	+3.53	0.55	1/4 3/4	+0.02 (0.90)
OLMo-3-7B	+0.03	+0.03	+0.14	+0.22	0.35	4/8 3/8	-0.02 (0.55)
OLMo-3-32B	+0.01	+0.03	+0.02	+0.09	none by 0.30	7/10 0/10	-0.24 (0.30)

Table 2: Per-capability pruning damage on the heterogeneous panel (nats/token, native tokenizer units; within-model differences). d_{math}^* is the largest measured density with $\Delta L_{\text{math}} \geq 1$; models are grouped by series (Qwen3, Gemma-3, Gemma-4, Muse, OLMo-3); “math@0.5” is ΔL_{math} at $d = 0.5$; “QA least | worst” counts, over the pre-cliff densities, how often QA is the least-damaged and how often the most-damaged capability the last column is the most negative QA response and the density at which it occurs. [†]prospective additions measured at four densities only. Generated by v51_panel_tables.py from prune_losses.json.

response is worth predicting; strength-only and per-configuration constants ask whether the source state helps; the no- D_0 variant asks what D_0 adds; A1, A2, and the continuous form ask whether the power structure adds anything. Errors are mean absolute errors in nats per token per capability, improvement is baseline MAE minus candidate MAE (positive means the candidate is better), bootstrap intervals are computed only on the leave-one-out panels with source states as clusters, and repeated probes, checkpoints of one run, and stages of one source are never counted as independent sources.

5 Results

We first establish what compression does to the capability vector on the heterogeneous panel (§5.1), then test prediction arm by arm with one template (inputs and baselines; source axis; configuration axis; boundaries) for pruning (§5.2), quantization (§5.3), and distillation (§5.4), and close with a synthesis and the limitations (§5.5). Mechanism evidence, recovery training, and the re-analysis of an earlier ability-space grid are supporting material (Appendix B).

5.1 Capability responses on the heterogeneous panel

Pruning. Table 2 gives per-capability damage at two densities, the density at which math damage first exceeds one nat, and how QA ranks before that cliff. Magnitudes differ by two orders across series at the same density (OLMo-3-32B +0.01 against Gemma-3-27B +4.0 on math at $d = 0.7$). The ordering of capabilities is series-specific: in the Qwen3 sizes and in OLMo-3-32B, QA is the least-damaged capability at every pre-cliff density and improves, by up to -0.79 nats at Qwen3-4B; in every Gemma-3 size, in Gemma-4-31B, and in Muse-30B, QA is the most-damaged capability at the mildest densities (Gemma-4-31B: +2.89 QA against +0.68 math at $d = 0.7$); OLMo-3-7B lies in between with differences of at most 0.15 nats. A noise screen sets the scale: the median $|\Delta L|$ at $d = 0.9$ is 0.013 nats, and 29 of the 62 nominal improvement cells across all arms exceed twice that median; the rest are treated as zero. Cliffs occur between $d = 0.5$ and 0.7 for Qwen3 and Gemma, at 0.55 for Muse-30B and 0.35 for OLMo-3-7B, and had not occurred by $d = 0.3$ for OLMo-3-32B. Robustness is not monotone in size within a series (Gemma-3-12B is the most robust Gemma-3 size; 27B

collapses earlier). We read this as heterogeneity a predictor has to reproduce, not as an invariant.

Quantization. Appendix Table 5 covers 14 models. Int8 and int6 stay within 0.01 nats of dense. Int4 damages small models (Gemma-3-270M +2.0, +2.4, +0.9) and barely touches large ones (Gemma-3-27B +0.33, +0.23, -0.67); a QA improvement under int4 recurs in five models and clears the noise screen. Int3 collapses 12 of the 14 models by 4.6 to 28 nats; OLMo-3-7B loses 2.8 to 4.7 and OLMo-3-32B only 0.4 to 1.2, so the collapse is common but not universal, and OLMo-3 is the robust series under both interventions. Within the tested bit-widths a learned exponential $q_c e^{-kb}$ fits held-out bits better than the fixed $q_c 4^{-b}$ shape (pooled MAE 6.7 against 7.6 nats), which says the fixed base is wrong for this quantizer and nothing more; the squared step-size ratio between 4 and 5 bits is $(7/15)^2 \approx 0.22$ (Appendix A.2), and a loss ratio would follow only under error-distribution assumptions we do not test.

Distillation. Black-box traces from two API teachers (600 per domain) train Gemma-3 students under a full-trace and an answer-only recipe in a paired design. The answer-only recipe hurts math more (+0.19 to +0.22 against +0.09 to +0.12 nats relative to the initial student) while QA improves under every recipe (-0.58 to -1.44); because the recipes differ in supervised tokens and update counts, this pairing does not isolate reasoning coverage, and the controlled comparisons are those of §5.4.

What the endpoint does and does not say. The same heterogeneity appears on independent secondary benchmarks for math and QA, where the primary-benchmark response predicts the secondary one better than any other capability’s response does, with intervals excluding zero in most hold-out protocols; for code the cross-capability predictor is as good or better (Appendix A.1, Fig. 3). A lower QA loss does not by itself mean better QA behavior: in the four pruned OLMo-3-32B cells whose QA loss lies below dense, exact-match accuracy does not rise. “QA improves” in this paper refers to the loss.

5.2 Pruning arm

Inputs, candidates, baselines. Inputs are N_0 , D_0 , and the dense capability loss $L_{0,c}$ of the source (Fig. 1, left); candidates and baselines are those of §4.2, fit on the nine Pythia development states.

Source axis (Table 3; Appendix Table 6). With the density fixed and one stage held out, the configuration-indicator regression reaches MAE 0.253 (math), 0.292 (code), and 0.806 (QA) against 0.413, 0.516, and 0.500 for the strongest simple baseline; the improvement interval excludes zero for code only, and for QA the regression is worse than predicting zero change. Same-input comparisons say which input carries the gain: adding the dense anchor $L_{0,c}$ to $\{N_0, D_0\}$ has intervals above zero for math (+0.55, [0.29, 1.00]) and code (+0.66, [0.38, 1.13]), whereas adding D_0 to $\{N_0, L_{0,c}\}$ does not (+0.36, [-0.04, 0.60]; +0.40, [-0.01, 0.63]); on an earlier split that also held out sizes, the D_0 interval cleared zero in one pruning cell (math) of six. On a frozen prospective at a new stage (step 96k on the 160M and 1.4B sources, between 64k and 143k), the frozen regression beat both its no- D_0 variant and the per-configuration median on all three capabilities, with the largest margin on math.

Configuration axis (Table 3; Appendix Table 7). On the development sources plus a new source (410M at 96k), pooled over the unseen densities 0.65 and 0.55, the frozen shared power form has MAE 0.371, 0.524, 0.759 against 1.37, 1.64, 1.61 for the strength-only curve: a source-blind curve cannot see that one state (160M at 143k) is several times more fragile than another. The same-input regression A2, added after the test, is better on math and code and within noise on QA, so the power structure adds nothing that the per-density regression lacks. Two frozen rounds on new sizes then set the range. For the 1B source, inside the development size range, at the interpolated densities over two stages, the power form, A2, and the source-free median curve are within 0.05 nats of one another on every capability; per stage (Appendix B.7) A2 is best at 112k on each capability and the median

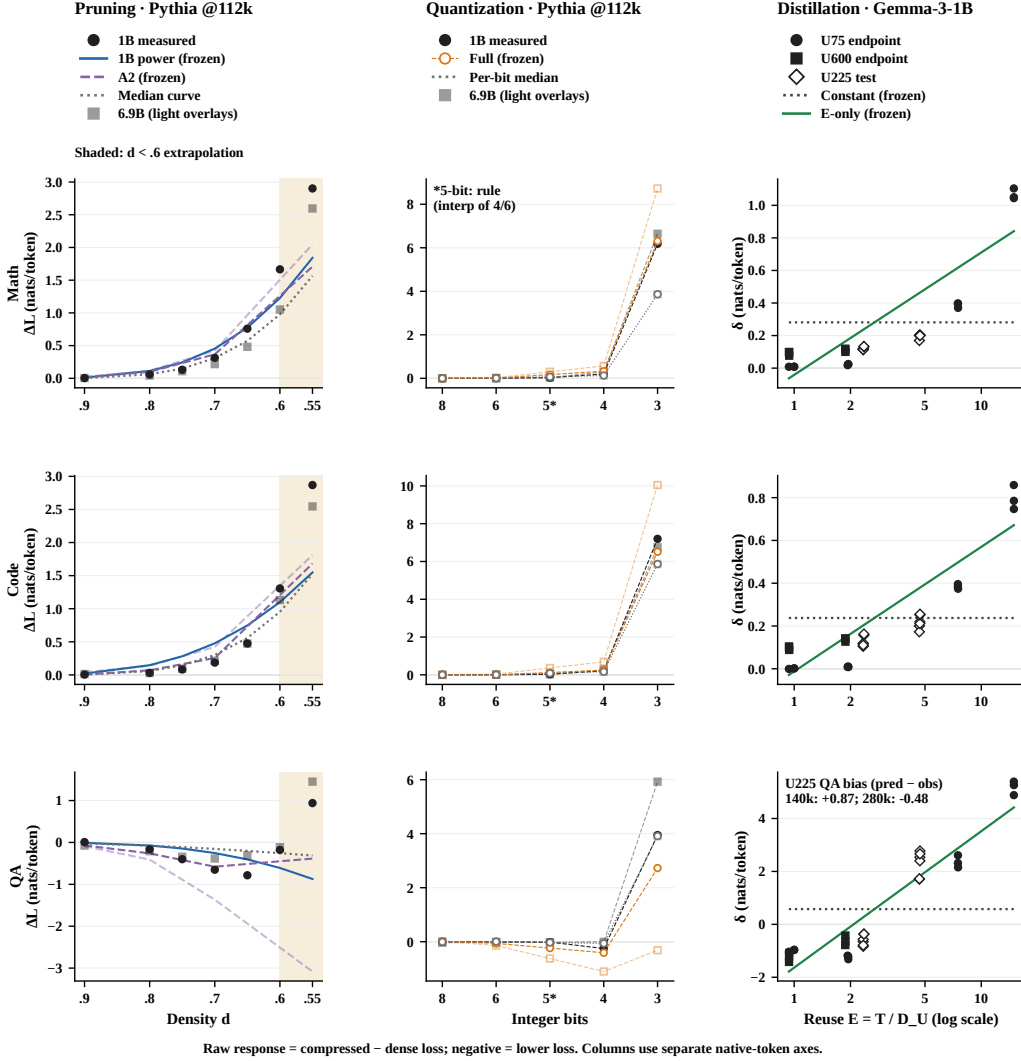


Figure 1: Measured capability responses and frozen predictions, one source per arm. Left: pruning of Pythia-1B at step 112k (filled circles) with the frozen shared power form, the per-density regression A2, and the source-free median curve; the 6.9B source at the same stage is overlaid in grey (measured squares and its A2 and median predictions) and the extrapolated density 0.55 is shaded. Middle: quantization of the same sources at the integer bit-widths (5-bit is the fixed interpolation rule); measured points are filled, the frozen configuration-indicator prediction is hollow, the per-bit median dotted; guide lines are visual only. Right: distillation of Gemma-3-1B, signed transfer δ_e against the reuse count E ; endpoint pools U75 and U600 (fit) and the unseen pool U225 (test), with the frozen constant and E -only forms and the QA bias of the E -only form at the two budgets. All values are native nats per token; columns use separate axes. Generated by plot_fig1_responses.py from the compare and run files listed in docs/FIGURE_BUILD_NOTES.md.

curve is best at 32k, driven by QA. For the 6.9B source, five times the largest development size, the source-conditioned forms lose to the median curve on math and QA and tie on code: the regression extends the panel’s size trend into QA improvements of one to three nats that do not occur. At density 0.55 every source-conditioned form fails on every source (errors 0.6 to 4.5 nats) and a source-free curve is closest; the single-stage 1B@96k prospective (Appendix Table 7) shows the same two facts. Under protocol B (early stages only; Appendix Table 9)

Test	unseen	origin	math	code	QA
Source axis: seen setting, unseen source state					
prune: fixed d , 3×3 , leave-one-stage-out	stage	LFRA	0.253/0.413 _{med}	0.292/0.516 _{med}	0.806/0.500 _o
quant: fixed b , 3×3 , leave-one-stage-out	stage	LFRA	1.144/1.410 _{med}	0.626/1.553 _{med}	1.989/1.473 _{med}
prune: stage 96k (160M, 1.4B), all d	stage	PFRA	0.090/0.343 _{nD}	0.128/0.179 _{nD}	0.426/0.438 _{nD}
quant: new stage 96k (160M/410M/1.4B), $b \geq 4$	stage	P-RA	0.029/0.084 _{pbm}	0.104/0.094 _{pbm}	0.173/0.134 _{pbm}
quant: new stage 96k (160M/410M/1.4B), int3	stage	P-RA	1.488/1.934 _{pbm}	1.963/0.843 _{pbm}	2.046/1.727 _{pbm}
distill: δ_c , 3×3 LoRA, leave-one-stage-out	stage	LFRA	0.014/0.026 _{nD}	0.057/0.044 _{nD}	0.263/0.159 _c
Configuration axis: unseen setting, size, stage, or pool					
prune: d 0.65/0.55, dev sources + 410M@96k	d	PRFA	0.371/0.283 _{A2}	0.524/0.279 _{A2}	0.759/0.818 _{A2}
prune: 1B@96k, $d = 0.65$	size, stage, d	PFFA	0.020/0.009 _{A1}	0.092/0.031 _{A1}	0.466/0.380 _{A2}
prune: 1B@96k, $d = 0.55$	size, stage, d	PFFA	0.930/0.273 _{so}	1.550/0.018 _{so}	1.426/0.530 _{so}
quant: 1B@96k, int4	size, stage	PFFA	0.116/0.068 _{pbm}	0.046/0.012 _{pbm}	0.059/0.054 _{nD}
quant: 1B@96k, int3	size, stage	PFFA	0.690/1.414 _{pbm}	0.318/0.001 _{pbm}	0.182/0.446 _{pbm}
prune: 1B@32k+112k, d 0.9–0.6	size, stage	PFFA	0.215/0.169 _{so}	0.191/0.119 _{A2}	0.261/0.227 _{med}
prune: 1B@32k+112k, $d = 0.55$	size, stage, d	PFFA	1.790/0.939 _{so}	1.488/0.089 _{so}	2.225/0.393 _{so}
quant: 1B@32k+112k, $b \geq 4$ (5-bit rule incl.)	size, stage, b	PFFA	0.064/0.020 _{pbm}	0.041/0.028 _{pbm}	0.157/0.052 _{pbm}
quant: 1B@32k+112k, int3	size, stage	PFFA	0.256/1.466 _{pba}	0.362/1.537 _{pba}	1.972/1.000 _{pbm}
prune: 6.9B@32k+112k, d 0.9–0.6	size $\uparrow$, stage	PFFA	0.299/0.120 _{med}	0.160/0.144 _{A2}	0.909/0.120 _{med}
prune: 6.9B@32k+112k, $d = 0.55$	size $\uparrow$, stage, d	PFFA	0.615/0.618 _{A2}	0.547/0.535 _{A2}	4.472/0.904 _{so}
quant: 6.9B@32k+112k, $b \geq 4$	size $\uparrow$, stage, b	PFFA	0.181/0.036 _{pbm}	0.177/0.043 _{pbm}	0.428/0.034 _o
quant: 6.9B@32k+112k, int3	size $\uparrow$, stage	PFFA	3.233/2.046 _{pba}	3.229/1.805 _{pba}	5.306/1.744 _{pba}
distill: unseen pool U225 (Gemma-3-1B)	pool	PFFA	0.191/0.122 _c	0.117/0.069 _c	0.706/0.710 _{TE}

Table 3: Prediction tests, all three arms (MAE in nats/token per capability, candidate / strongest baseline, bold = better). The candidate is the source-conditioned predictor of each arm (configuration-indicator regression on the source axis and for quantization; shared power form for pruning on the configuration axis; E -only form for the distillation pool test). The baseline shown is the stronger of the same-input and the simple baseline per capability, named by code: nD = no- D_0 variant, med = per-configuration median or median development curve, 0 = zero change, so = strength-only curve, A1 = $\gamma=1$ power form, A2 = per-density regression with fixed interpolation, ct = continuous two-term form, pbm/pba = per-bit median/mean, c = constant, TE = joint $T+E$ form. “unseen” lists what the test holds out (size $\uparrow$ = five times beyond the development range). Origin code, four letters: prediction (P frozen prospective, L leave-one-out), same-input baseline, simple baseline (F pre-specified with the candidate, R computed after the test, – none), selection (A: all pre-specified forms reported). Improvements, both baselines separately, and bootstrap intervals are in Appendix Tables 6–8. Generated by v52_prediction_tables.py.

the 6.9B interpolation errors shrink to the level of the source-free curves without beating them.

Deliverable predictor and confirmation panel (Table 10). After the frozen rounds we refit the candidates on an expanded development set of 17 Pythia states that includes the unblinded 1B and 6.9B checkpoints (the earlier frozen predictions stay as recorded) and chose one deliverable by a pre-committed leave-one-source-out rule: the per-density regression A2 and the shared power form tie within 0.02 nats (0.479 against 0.494), and the rule resolves the tie toward the continuous five-parameter form that vanishes at $d = 1$. Its predictions for three checkpoints that had never entered any fit (410M@48k, 1.4B@112k, 6.9B@80k) at three densities between the development grid points were committed before measurement. On the nine cells the selected form beats the strength-only curve and zero change on every checkpoint, has the lowest math error of all predictors (A2 is near-exact on math at 1.4B@112k), ties the median development curve on code, and loses QA to every source-free curve by 0.3 to 0.6 nats because it over-predicts the QA improvement. The source-free median curve therefore has the lowest aggregate error (0.237 against 0.386), and the deliverable is stated per capability: a compact source-conditioned relation for math and code inside the fitted range, a source-free curve for QA. No universality is claimed for the fitted exponents.

Boundaries. Source-state information improves pruning prediction at seen densities and at interpolated densities for sources inside the development size and stage range, mostly for math and code, and does not require the power structure. It does not extend to a source five times larger, to density 0.55, or reliably to QA, whose response the current inputs do not carry.

5.3 Quantization arm

Inputs, candidates, baselines. The same inputs; the configuration-indicator regression per bit-width against its no- D_0 variant, the per-bit mean and median, and zero change (Fig. 1, middle).

Source axis. Pooled over the four bit-widths with a stage held out, the regression improves on the per-bit median by +0.27 (math), +0.93 (code, interval excluding zero), and -0.52 (QA); adding D_0 has an interval above zero for code (+1.45, [0.15, 2.10]), a positive point estimate for math, and none for QA. The pooled number is dominated by int3, so the frozen step-96k test on three new sources is reported by regime: at 4 bits and above the responses are near zero and candidate and median errors are both small (0.03 against 0.08, 0.10 against 0.09, 0.17 against 0.13), which demonstrates no gain at that scale and does not show that none exists; at int3 the candidate is better for math (+0.45) and worse for code (-1.12) and QA (-0.32), so the collapse magnitude does not transfer consistently across capabilities.

Configuration axis (Appendix Table 8). The tested settings are the integer bit-widths, so the unseen-configuration tests are the 5-bit rule and the unseen sizes. For the in-range 1B source the regression is worse than the per-bit median at 4 bits and above, and at int3 it is better for math and code (+1.2 nats each) and worse for QA (-1.0). For the 6.9B source it is worse everywhere, by 0.1 to 0.4 nats above 3 bits and by 1.2 to 3.6 nats at int3, where it overshoots the collapse. Under protocol B (Appendix Table 9) the 6.9B differences above 3 bits shrink to at most 0.05 nats and at int3 the regression is ahead at 112k (1.29 against 3.8 to 4.1) and behind at 32k. The single-stage 1B@96k test had the regression ahead on the int3 aggregate (0.40 against 0.62 for the median and 2.46 without D_0) with the same math-positive, code-negative pattern.

Group size as a second configuration axis (Table 12). Bit-width alone leaves the near-zero and collapse regimes with little in between, so we added the group size g of a grouped symmetric round-to-nearest quantizer (per-group absmax scale; g equal to the row width reproduces the per-channel quantizer exactly). On six development states ($b \in \{3, 5\}$, $g \in \{64, 256\}$) the responses have structure: 5-bit cells stay within 0.1 nats, 3-bit damage grows steeply with training stage (late 160M collapses by 15 to 17 nats, late 1.4B loses 0.6 to 2.2)

and roughly halves when g drops from 256 to 64. Three predictors were frozen on those 24 cells and tested on unseen configurations: at the unseen bit-width $b = 4$ a low-order two-dimensional form in $(\log_2 q_{\max}, \log_2 g)$ with source-state coefficients has MAE 0.19, 0.21, 0.44 against 0.45, 0.53, 0.48 for zero change and 0.55, 0.73, 0.54 for the per-configuration median, and at the unseen group size $g = 128$ it has 0.22, 0.32, 0.99 against 1.24, 1.35, 1.24 for zero change, on all three capabilities. The compact separable candidate $A_c(\mathbf{x}) q_{\max}^{-p_c}(g/g_{\text{ref}})^{q_c}$ is worse than zero change on the bit test, so the response does not factor into a state amplitude and a shared shape, and the same-input bilinear interpolation fails outside the development corners. These successes are configuration transfer on seen source states; on the one new state tested (1B@96k, three cells) the two-dimensional form is best on code and behind the source-free curves on math and QA.

Boundaries. In the near-zero regime no predictor is distinguishable from a per-bit constant on these panels. In the collapse regime the source-conditioned regression carries information for in-range sizes, with a sign that differs by capability, and does not transfer to a five-times larger source. Which statement holds depends on the fitting protocol, so a quantization result here is always stated with its precision regime and protocol.

5.4 Distillation arm

Inputs, candidates, baselines. The response is the signed transfer δ_c relative to the initial student. On the source axis the inputs are the student’s N_0 , D_0 , and $L_{0,c}$ with a fixed teacher and pool; on the pool axis they are the processed tokens T and the reuse count $E = T/D_U$ of a fixed student (Fig. 1, right).

Source axis. With one stage held out on the nine-state LoRA panel, the linear model in $\{N_0, L_{0,c}, D_0\}$ halves the error of a constant for math (0.014 against 0.031), ties for code (0.057 against 0.061, with the no- D_0 variant at 0.044), and is worse than a constant for QA (0.263 against 0.159); nine cells give no usable interval.

Configuration axis. Four forms frozen on the endpoint pools U75 and U600 were applied to an unseen middle pool U225, whose reuse count (2.3 to 4.7) lies inside the endpoint range; because the endpoints are pool prefixes and U225 is randomly sampled, the sampling protocol changes as well. The reuse-count form beats the constant only for QA (0.706 against 1.449) and ties the joint form; for math and code the constant is best (0.122, 0.069) and the fitted forms over-extrapolate. All four forms are reported; the per-capability recommendation (reuse count for QA, constant for math and code) is a reading made after the result. The residual is dominated by systematic bias: the QA reuse-count prediction is -0.87 nats off at one budget and $+0.48$ at the other, a form misfit, and around that bias the variation across pool samples exceeds the variation across training seeds (QA 0.43 against 0.04 nats), which describes this design of three pools and two seeds and decomposes no population variance.

Boundaries. A transfer predictor from the student state exists for math on the controlled panel; for QA a constant is better on both axes; the reuse-count coordinate carries information for QA on the unseen pool with a budget-dependent bias. Both panels are small, the pool test changes two things at once, and the absolute-exposure matrix that separates them lies outside the data cut-off of this paper.

5.5 Cross-arm synthesis and limitations

Inputs and forms (Appendix Fig. 14). The dense capability vector is the input with the clearest support: adding it has intervals above zero for math and code on the controlled pruning panel, and every successful frozen test uses it. Training tokens add information beyond $\{N_0, L_{0,c}\}$ with interval support in one cell (quantization, code) and positive point estimates in most others, and never for QA. On the same inputs the shared power form never beats the per-density regression with a fixed interpolation rule, so no exponent is established by these data.

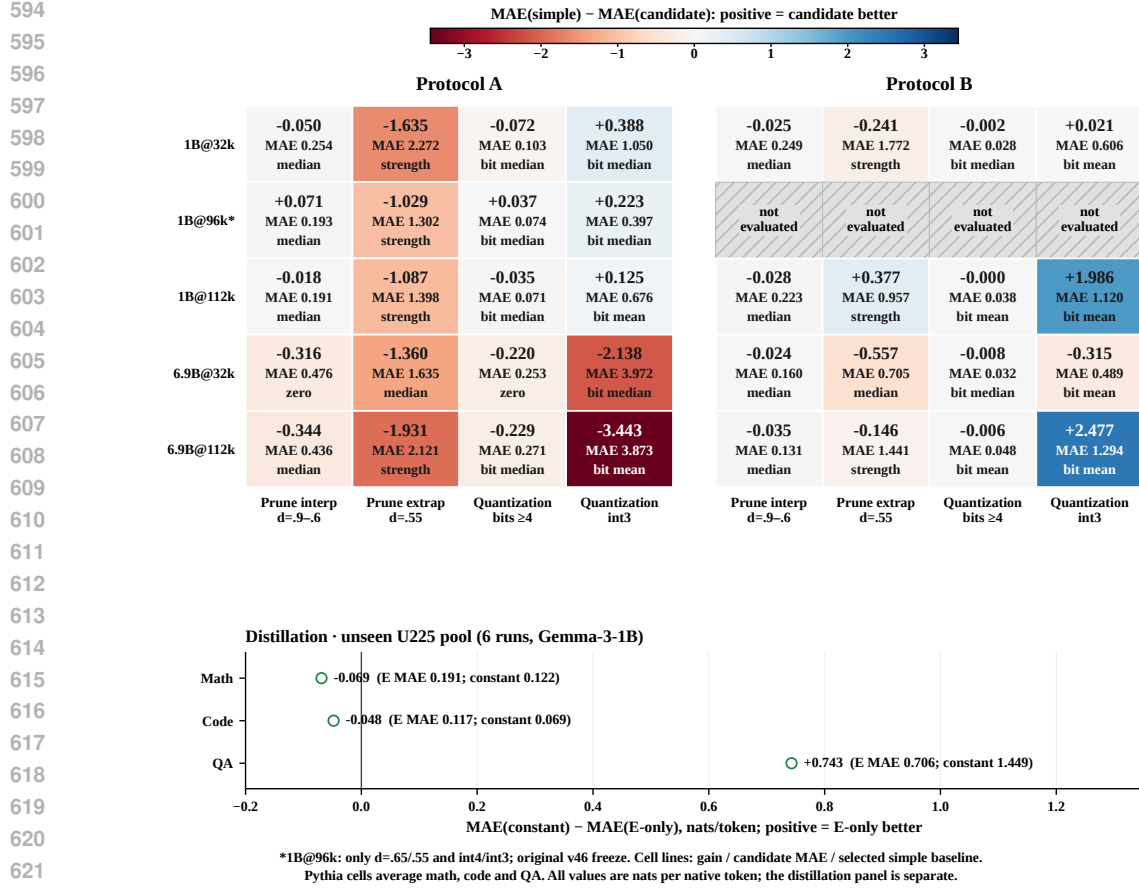


Figure 2: Where transfer holds and where it fails. Top: improvement of the headline candidate (shared power form for pruning, configuration-indicator regression for quantization) over the strongest source-free baseline, MAE difference averaged over the three capabilities, for each new source (rows), regime (columns), and fitting protocol (panels); each cell also prints the candidate MAE and the baseline that was strongest. Positive (blue) means the candidate is better. The 1B@96k row is the earlier single-stage freeze, evaluated at $d = 0.65/0.55$ and int4/int3 only; its protocol-B cells were not evaluated. Bottom: the distillation pool test, E -only form against the constant on the unseen pool U225, per capability. Generated by `plot_fig3_transfer.py`.

Which transfers were tested (Fig. 2). Unseen stage inside the range: supported for pruning (leave-one-out and frozen). Unseen density inside the fitted range: supported on the development sources; on par with source-free curves at a new in-range size. Density 0.55: fails for every source-conditioned form. Size five times beyond the range: fails in both arms. Unseen bit-width: the 5-bit rule of the per-channel predictor is worse than a per-bit constant, but with group size as a second axis a low-order form predicts the unseen 4-bit and the unseen group size on seen states. Unseen density between grid points on three new checkpoints: supported for math and code, not for QA. Unseen pool: reuse count helps QA only. Each is one or two source states or one pool; none is a population statement.

Resource-ratio models. A method-agnostic response in the storage or active-parameter ratio was tested on the heterogeneous panel: at the common storage ratio 0.5, pruning to $d = 0.5$ costs median +5.9, +8.2, and +2.9 nats while int8 quantization costs none; over 75 storage-matched pairs (14 with both sides pre-cliff) the mean absolute method gap is 9.3 nats with sign agreement 0.61, and the two shared-law hypotheses fit on pre-cliff cells reach held-out sign accuracies of 0.51 and 0.37. These models are rejected; whether another

single form could unify the arms is untested. A capability- and cost-aware method selection would use predictors like these as loss constraints; an offline replay with frozen predictions quantifies the decision cost of prediction error, but no held-out selection validation was run, so nothing here is a validated selector.

Limitations. The controlled panels have three sizes, three stages, and three pools, so intervals are conditional on the panel. QA is the capability the current inputs carry worst on every axis. The quantization conclusions depend on the precision regime and the fitting protocol. The distillation pool test changes sampling protocol and pool size together. Code specificity of the endpoint was not detected at the available resolution. Recovery training, geometry, and the shared-law tests rest on single models or pre-cliff subsets and are supporting evidence only.

6 Conclusion

We asked whether the model state observable before compression, together with the intervention setting, predicts the capability-conditioned loss after pruning, quantization, and distillation, and over which ranges such predictions transfer. The answer has three parts. Responses are heterogeneous across capabilities and model series in sign and magnitude, so a capability-conditioned endpoint is necessary and no single ordering holds. Source-state information helps: the dense capability vector carries the prediction, source-conditioned pruning predictors beat source-free curves at seen densities and at interpolated densities on the development sources, mostly for math and code, and match them at a new in-range size; the power structure adds nothing over a per-density regression, and training tokens add little beyond the dense anchor. The transfer range is bounded: the same predictors fail at density 0.55 and at a source five times larger, quantization predictors add value only in the collapse regime and under some fitting protocols, unless the group size is added as a second axis, where a low-order relation predicts unseen configurations on seen states, and for distillation a reuse-count coordinate helps QA on an unseen pool while a constant is best for math and code. Method-agnostic resource-ratio laws fail a matched-storage test. These results identify conditional predictive relationships and their tested limits and provide the measurements, registers, and comparison protocol on which capability-conditioned scaling-down laws can be developed; widening the size range of the development panel and finding inputs that carry the QA response are the next steps, and both need new experiments.

References

- Tamay Besiroglu, Ege Erdil, Matthew Barnett, and Josh You. Chinchilla scaling: A replication attempt. arXiv preprint arXiv:2404.10102, 2024.
- Akshita Bhagia, Jiacheng Liu, Alexander Wettig, David Heineman, Oyvind Tafjord, Ananya Harsh Jha, Luca Soldaini, Noah A. Smith, Dirk Groeneveld, Pang Wei Koh, Jesse Dodge, and Hannaneh Hajishirzi. Establishing task scaling laws via compute-efficient model ladders. In Conference on Language Modeling (COLM), 2025.
- Dan Busbridge, Amitis Shidani, Floris Weers, Jason Ramapuram, Etai Littwin, and Russ Webb. Distillation scaling laws. In International Conference on Machine Learning, 2025.
- Xiaodong Chen, Yuxuan Hu, Xiaokang Zhang, Yanling Wang, Cuiping Li, Hong Chen, and Jing Zhang. P² law: Scaling law for post-training after model pruning. In Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics, 2025.
- Tim Dettmers and Luke Zettlemoyer. The case for 4-bit precision: k-bit inference scaling laws. In International Conference on Machine Learning, 2023.
- Elias Frantar and Dan Alistarh. Sparsegpt: Massive language models can be accurately pruned in one-shot. In International Conference on Machine Learning, 2023.
- Elias Frantar, Carlos Riquelme, Neil Houlsby, Dan Alistarh, and Utku Evci. Scaling laws for sparsely-connected foundation models. In International Conference on Learning Representations, 2024.

- Samir Yitzhak Gadre, Georgios Smyrnis, Vaishaal Shankar, Suchin Gururangan, Mitchell Wortsman, Rulin Shao, Jean Mercat, Alex Fang, Jeffrey Li, Sedrick Keh, Rui Xin, Marianna Nezhurina, Igor Vasiljevic, Luca Soldaini, Jenia Jitsev, Alex Dimakis, Gabriel Ilharco, Pang Wei Koh, Shuran Song, Thomas Kollar, Yair Carmon, Achal Dave, Reinhard Heckel, Niklas Muennighoff, and Ludwig Schmidt. Language models scale reliably with over-training and on downstream tasks. In International Conference on Learning Representations, 2025.
- Andrei Ghitu, Nirav Desai, and Alex Boier. Scaling laws for task-specific LLM distillation. arXiv preprint arXiv:2606.24747, 2026.
- Guy Gur-Ari, Daniel A. Roberts, and Ethan Dyer. Gradient descent happens in a tiny subspace. arXiv preprint arXiv:1812.04754, 2018.
- Babak Hassibi and David G. Stork. Second order derivatives for network pruning: Optimal brain surgeon. In Advances in Neural Information Processing Systems, 1992.
- Gabriel Ilharco, Marco Tulio Ribeiro, Mitchell Wortsman, Suchin Gururangan, Ludwig Schmidt, Hannaneh Hajishirzi, and Ali Farhadi. Editing models with task arithmetic. In International Conference on Learning Representations, 2023.
- Ajay Jaiswal, Zhe Gan, Xianzhi Du, Bowen Zhang, Zhangyang Wang, and Yinfei Yang. Compressing llms: The truth is rarely pure and never simple. In International Conference on Learning Representations, 2024.
- Tanishq Kumar, Zachary Ankner, Benjamin F. Spector, Blake Bordelon, Niklas Muennighoff, Mansheej Paul, Cengiz Pehlevan, Christopher Ré, and Aditi Raghunathan. Scaling laws for precision. In International Conference on Learning Representations, 2025.
- Yann LeCun, John S. Denker, and Sara A. Solla. Optimal brain damage. In Advances in Neural Information Processing Systems, 1989.
- Margaret Li, Sneha Kudugunta, and Luke Zettlemoyer. (mis)fitting: A survey of scaling laws. In International Conference on Learning Representations, 2025.
- Ruikang Liu, Yuxuan Sun, Manyi Zhang, Haoli Bai, Xianzhi Yu, Tiezheng Yu, Chun Yuan, and Lu Hou. Quantization hurts reasoning? an empirical study on quantized reasoning models. In Conference on Language Modeling (COLM), 2025.
- Nicholas Lourie, Michael Y. Hu, and Kyunghyun Cho. Scaling laws are unreliable for downstream tasks: A reality check. In Findings of the Association for Computational Linguistics: EMNLP 2025, 2025.
- Eric J. Michaud, Ziming Liu, Uzay Girit, and Max Tegmark. The quantization model of neural scaling. In Advances in Neural Information Processing Systems, 2023.
- David Owen. How predictable is language model benchmark performance? arXiv preprint arXiv:2401.04757, 2024.
- Andrei Panferov, Alexandra Volkova, Ionut-Vlad Modoranu, Vage Egiazarian, Mher Safaryan, and Dan Alistarh. Unified scaling laws for compressed representations. In Advances in Neural Information Processing Systems, 2025.
- Yangjun Ruan, Chris J. Maddison, and Tatsunori Hashimoto. Observational scaling laws and the predictability of language model performance. In Advances in Neural Information Processing Systems, 2024.
- Rylan Schaeffer, Brando Miranda, and Sanmi Koyejo. Are emergent abilities of large language models a mirage? In Advances in Neural Information Processing Systems, 2023.
- Rylan Schaeffer, Hailey Schoelkopf, Brando Miranda, Gabriel Mukobi, Varun Madan, Adam Ibrahim, Herbie Bradley, Stella Biderman, and Sanmi Koyejo. Why has predicting downstream capabilities of frontier ai models with scale remained elusive? In International Conference on Machine Learning, 2025.

Ayan Sengupta, Siddhant Chaudhary, and Tanmoy Chakraborty. Pruning laws for large language models. arXiv preprint arXiv:2504.04342, 2025. v2, EMNLP 2026; v1 (2025) was titled Compression Laws for Large Language Models.

Pratyusha Sharma, Jordan T. Ash, and Dipendra Misra. The truth is in there: Improving reasoning in language models with layer-selective rank reduction. In International Conference on Learning Representations, 2024.

Ibne Farabi Shihab, Sanjeda Akter, and Anuj Sharma. Beyond variance: Knowledge-aware LLM compression via fisher-aligned subspace diagnostics. arXiv preprint arXiv:2601.07197, 2026.

Mingjie Sun, Zhuang Liu, Anna Bair, and J. Zico Kolter. A simple and effective pruning approach for large language models. In International Conference on Learning Representations, 2024.

Lu Yin, Ajay Jaiswal, Shiwei Liu, Souvik Kundu, and Zhangyang Wang. Junk DNA hypothesis: Pruning small pre-trained weights irreversibly and monotonically impairs “difficult” downstream tasks in LLMs. In International Conference on Machine Learning, 2024.

Jiawei Zhao, Zhenyu Zhang, Beidi Chen, Zhangyang Wang, Anima Anandkumar, and Yuan-dong Tian. Galore: Memory-efficient llm training by gradient low-rank projection. In International Conference on Machine Learning, 2024.

Chenxi Zhou, Pengfei Cao, Jiang Li, Bohan Yu, Jinyu Ye, Jun Zhao, and Kang Liu. Task-stratified knowledge scaling laws for post-training quantized large language models. In Findings of the Association for Computational Linguistics: ACL 2026, 2026.

A Appendix

A.1 Probes and the capability-conditioned loss

Each capability is measured on a fixed probe set of prompt–reference pairs drawn once, with a fixed seed, from held-out benchmark splits that are disjoint from each training and recovery source used in the paper. The primary measurement benchmarks are MATH-500 for mathematical reasoning, MBPP for code generation, and 2WikiMultiHopQA for multi-hop question answering. For each capability we draw 128 probes and split them by index parity into an estimation half, used only for gradient and Fisher statistics, and a measurement half of 64 probes on which each reported loss is computed, so that mechanism statistics and evaluations do not share sampling noise. Secondary benchmarks used for the measurement checks below are GSM8K, HumanEval, and HotpotQA on the pruning panel (no training involved) and SVAMP, HumanEval, and TriviaQA where distilled models are scored, chosen to exclude the benchmarks from which the distillation traces were derived; a sample-level audit found no overlap between any measurement probe and any training pool. The measurement half is 64 probes per capability for every result in the paper except where a 512-probe re-measurement is stated.

Capability specificity as a prediction test. On a six-model, five-density pruning panel (d from 0.9 to 0.7), the response on the primary benchmark of each capability was used to predict the response on the secondary benchmark of the same capability through a two-parameter affine map fit out of fold, and compared with the best cross-capability predictor chosen the same way. With whole models held out, the same-capability gain is +0.04 nats for math (95% model-bootstrap interval [0.03, 0.05]) and +0.14 for QA ([0.02, 0.27]); with whole densities held out it is +0.14 ([0.03, 0.30]) and +0.21 ([0.02, 0.52]). For code the cross-capability predictor is as good or better (−0.06, [−0.16, −0.00] with densities held out), so code specificity is not established at this resolution (Fig. 3). Six model clusters give descriptive intervals only.

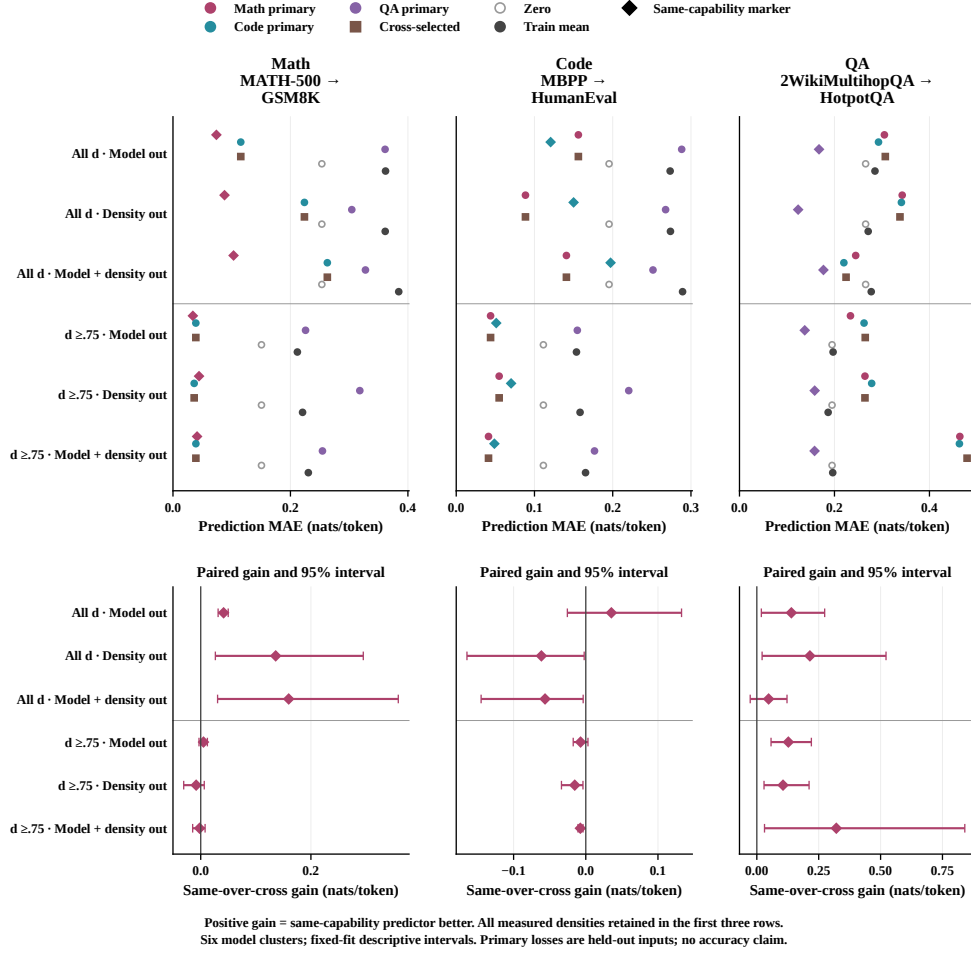


Figure 3: Measurement support for capability specificity. Top: out-of-fold MAE of predicting the secondary-benchmark response of each capability from the primary-benchmark response of the same capability (diamond) or of another capability, against zero and train-mean controls, on the six-model, five-density pruning panel; rows are hold-out protocols with all densities and with $d \geq 0.75$ only. Bottom: paired same-over-cross gain with 95% model-bootstrap intervals (six clusters, fixed out-of-fold fits). Generated by plot_fig4_measurement.py from v26-loss-validity-pred/summary.json.

Loss and behavior. Loss-to-accuracy links were fit on GSM8K accuracy, MBPP pass@1, and TriviaQA exact match and token F1 across seven pruned models; all four outcomes are compatible with one cross-family sigmoid in the loss, with no resolved family-specific term. A lower QA loss does not by itself imply better QA behavior: the four pruned OLMo-3-32B cells whose QA loss lies below dense show no increase in exact match. This audit covers pruning only.

Reference conditions. For a subset of models the direct-answer and given-reasoning-answer conditional distributions were measured in addition to the full-solution reference; they differ from it for math and QA and coincide for code, where the reference is the function body, so the numbers in this paper are specific to the full-solution reference and are not evidence of format robustness.

The loss is the conditional cross-entropy of the reference completion given the prompt, summed over completion tokens and divided by the number of completion tokens, as in

Eq. equation 2. Prompts and references are tokenized with the model’s own tokenizer and truncated to 1024 tokens; prompt tokens are masked out of the loss. All within-model responses ΔL_c are differences in native-token nats on identical token sequences. Because native-token units differ across tokenizers, cross-family comparisons of absolute levels use per-byte normalization or are reported separately, and no cross-tokenizer quantity is treated as a common numerical scale. Negative responses are retained as measured; nothing is clipped at zero.

A.2 Compression protocols

Pruning. Global magnitude pruning removes the smallest weights across all transformer matrices (attention and MLP projections; embeddings, output head, biases, and norms are excluded) to a target retained density d . The global threshold is estimated from a seeded sample of absolute weights so that the same procedure scales to 30B-parameter models; each density is applied to the same dense reference weights, which are held on the CPU for the largest models so that a single 80 GB device suffices.

Quantization. Weights are fake-quantized per output channel with symmetric round-to-nearest at bit-width b , using the integer range $[-(2^{b-1} - 1), 2^{b-1} - 1]$ and a per-channel scale set by the channel’s maximum absolute value; activations are left in bfloat16. The dense weights are restored between bit-widths. The quantizer is fake `quantize_per_output_channel` in the released code: $q_{\max} = 2^{b-1} - 1$ is the largest positive integer, integers are clamped to $[-q_{\max}, q_{\max}]$ ($2^b - 1$ values), and the scale is the channel’s maximum absolute weight divided by $q_{\max}$. With a fixed clipping range the step size from 4 to 5 bits shrinks by 7/15, so the squared step ratio is $(7/15)^2 \approx 0.22$; in general the squared step ratio between adjacent bit-widths is $(q_{\max}(b)/q_{\max}(b+1))^2$, not a constant 1/4. A squared step ratio becomes a loss ratio only under assumptions about the error distribution and the local response that we do not test, so the fixed 4^{-b} shape is one candidate among others rather than a derived reference. Nominal storage ratios are computed from bit-widths; no memory or latency was measured.

Distillation. Students are fine-tuned on black-box teacher traces (a frontier API teacher answering GSM8K, HotpotQA, and CodeAlpaca prompts; 600 traces per domain) under the full-trace recipe, with strict LoRA adapters of rank 16 and scaling 32 on the query, key, value, output, gate, up, and down projections, AdamW at learning rate 10^{-4} , an effective batch of 16 sequences, and a cosine schedule with 3% warmup. A run aborts if the adapter library is unavailable and does not fall back to full fine-tuning silently, and the actual training mode and trainable parameter count are recorded in each manifest. Token accounting distinguishes processed tokens (all input tokens seen, including repetitions), unique pool tokens, and supervised completion tokens. The data pool of U examples per domain is either the first U rows of the trace file (the historical protocol) or a random sample drawn with a pool-sampling seed that is independent of the training seed; pools are identified by the sorted hashes of their samples and pairwise overlaps are reported. The historical schedule spanned a fixed number of epochs, so its cosine horizon varied with pool size; the controlled matrix of §4.2 uses a schedule defined on the same absolute processed-token budget for each pool, and the two protocols are kept apart in fitting.

A.3 The controlled Pythia panel

The panel uses the public Pythia checkpoints at sizes 160M, 410M, and 1.4B and pretraining steps 16,000, 64,000, and 143,000, which correspond to 33.6B, 134.2B, and 299.9B processed tokens at the published batch of 1024×2048 tokens per step. The size coordinate N_0 counts transformer-matrix parameters only (the query, key, value, output, and MLP matrices of each layer), computed from the architecture configuration, and the training-tokens coordinate is $D_0 = \text{step} \times 2,097,152$. Each checkpoint’s weight identity was verified before use: the 2.8B revisions published for different steps were found to load identical weights and were excluded, and an integrity guard now rejects any pair of nominally different checkpoints with identical measured panels. New sources added for the prospective tests (Pythia-1B at

Cohort	Sources	Interventions	Probes	Role
Heterogeneous panel	12 dense checkpoints, 5 series from 4 families: Qwen3 0.6B/1.7B/4B, Gemma-3 270M–27B, Gemma-4-31B, Muse-30B, OLMo-3 7B/32B (+2 prospective: Qwen3 8B/14B)	global magnitude pruning, 10 densities in [0.3, 0.9]; RTN int8/6/5/4/3	64 per capability, primary benchmarks	phenomena; cross-family transfer; noise screen
Controlled Pythia panel	sizes 160M/410M/1.4B $\times$ steps 16k/64k/143k (9 states); new sources 160M, 410M, 1.4B at 96k; 1B at 32k/96k/112k; 6.9B at 32k/112k	pruning: 4 dev densities 0.9–0.6, tests at 0.75/0.65/0.55; bits 8/6/4/3, test 5	same	development set; source-axis and configuration-axis tests
Distillation, controlled	the 9 Pythia states as students; fixed teacher pool (600 traces per domain); LoRA	fixed recipe	same	δ_c source transfer
Distillation, pools	Gemma-3-1B student; pools U75 and U600 (fit), U225 (test); 3 pool seeds $\times$ 2 training seeds	processed tokens T ; reuse count $E = T/D_U$	same	configuration axis (pool)
Distillation, heterogeneous	Gemma-3 270M–12B, OLMo-3-7B, Qwen3 0.6B–4B students; two API teachers; full-trace vs answer-only	recipe	same	phenomena only

Table 4: Cohorts and their roles. Counts are generated from the result files (v51_panel_tables.py); the two Qwen3 prospectives were measured at four densities only. The 2.8B Pythia checkpoints were excluded because the step revisions we downloaded loaded identical weights (hash check); nothing is claimed about other revisions.

steps 32,000, 96,000, and 112,000; Pythia-6.9B at 32,000 and 112,000) were confirmed to have not entered any fit or model selection before their predictions were frozen.

Model	Series	ΔL_c at int4			ΔL_c at int3			bits
		math	code	QA	math	code	QA	
Qwen3-0.6B	Qwen3	+0.56	+0.73	+0.21	+8.83	+10.83	+6.61	8,6,5,4,3
Qwen3-1.7B	Qwen3	+0.40	+0.16	-0.21	+9.53	+9.46	+8.10	8,6,5,4,3
Qwen3-4B	Qwen3	+0.13	+0.19	+0.39	+10.73	+11.85	+5.99	8,6,5,4,3
Qwen3-8B [†]	Qwen3	+0.28	+0.48	-0.80	+12.86	+13.55	+9.45	8,6,5,4,3
Qwen3-14B [†]	Qwen3	+0.09	+0.31	-0.23	+9.30	+10.79	+6.45	8,6,4,3
Gemma-3-270M	Gemma-3	+1.99	+2.35	+0.88	+26.15	+27.94	+25.93	8,6,5,4,3
Gemma-3-1B	Gemma-3	+0.95	+0.59	+0.88	+15.33	+14.12	+9.80	8,6,5,4,3
Gemma-3-4B	Gemma-3	+0.72	+0.50	-0.27	+21.73	+19.24	+15.30	8,6,5,4,3
Gemma-3-12B	Gemma-3	+0.40	+0.32	-0.42	+20.19	+22.92	+20.75	8,6,5,4,3
Gemma-3-27B	Gemma-3	+0.33	+0.23	-0.67	+18.46	+17.15	+16.10	8,6,5,4,3
Gemma-4-31B	Gemma-4	+0.94	+0.58	+1.28	+20.35	+19.85	+14.90	8,6,5,4,3
Muse-30B	Muse	+1.58	+2.06	-0.11	+11.14	+11.18	+8.36	8,6,5,4,3
OLMo-3-7B	OLMo-3	+0.16	+0.21	-0.07	+4.70	+4.63	+2.76	8,6,5,4,3
OLMo-3-32B	OLMo-3	+0.04	+0.06	+0.02	+0.68	+1.22	+0.38	8,6,5,4,3

Table 5: Per-capability quantization damage (per-output-channel symmetric RTN, weights only; nats/token). int8 and int6 are within 0.01 nats of dense for every listed model and are omitted. [†]prospective additions. Generated by v51_panel_tables.py from quant_losses.json.

A.4 Registration, freezing, and labels

Each prediction test in Table ?? follows one order. The functional forms, the development set, and each standardization and coefficient are fixed and committed first. For a new source the dense capability loss is then measured, since it is a permitted pre-compression input; the predictions for each configuration and candidate are written to a versioned file and committed to the repository, with the commit hash recorded; and only then is the compressed or distilled outcome measured. Information budgets are reported separately: K0 uses pre-compression information only, K1 adds one pre-specified target-side calibration point, and post-hoc rescaling on target outcomes is reported only as an oracle diagnostic. Each comparison carries the four-part origin code defined with Tables 6–8; the label of a frozen prediction never depends on its outcome. Register files (paths in the repository; content hashes in docs/MANUSCRIPT_REWRITE_MAP.md): v36b-input-comparison/summary.json (leave-one-step-out source axis, both arms), v38-prospective/register.json (frozen step-96k coefficients), v40-prune-strength/register.json (shared power form and source-free curves), v41-distill-newpool/summary.json (the four frozen pool forms), v46-p1-newsource/predictions_frozen.json (1B@96k), v49-p1v2/register.json and predictions_*.json (1B and 6.9B pairs, protocols A and B). When a diagnostic overturns an earlier headline, the earlier claim is downgraded in the text and the prediction itself stays as committed.

A.5 Reproduction

All tables are generated from machine-readable result files by CPU-only scripts that run in seconds and require neither model weights nor a GPU; the repository mirrors the scripts and the result files they read, and the main prediction table is produced by a single command. Model measurements are reproducible from the recorded revisions, probe seeds, and protocols above.

B Supporting analyses

B.1 Capabilities occupy distinct regions of gradient space

Convergent validity (block structure). For each of nine benchmarks we measure a gradient signature (the benchmark-conditional diagonal Fisher) and compare all pairs. After removing the component shared across benchmarks, signatures of same-capability benchmarks

are positively aligned (cosine $+0.36$ to $+0.45$) while signatures of different capabilities are anti-aligned (-0.23 to -0.25), in each of the ten models from five series for which the block analysis was run (Figs. 4–13); the C4 control sits between blocks. (preliminary evidence: point estimates; example-bootstrap, permutation, and prompt-stability tests are left to future work) The mathematics benchmarks cluster regardless of format, and within-block fine structure matches intuition (SVAMP, grade-school word problems, is the loosest member of the math block). A projected-gradient SVD version gives the same verdict in subspace terms: principal-angle similarity within capability exceeds cross-capability similarity by $2\text{--}4\times$.

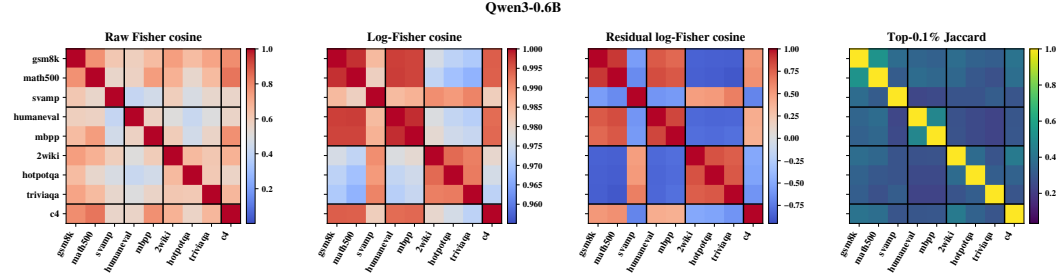


Figure 4: Capability regions in gradient space for Qwen3-0.6B: pairwise similarity of per-benchmark gradient signatures under four metrics (raw Fisher cosine, log-Fisher cosine, shared-component-removed log-Fisher cosine, top-0.1% coordinate Jaccard). Black lines separate the math, code, QA, and control blocks.

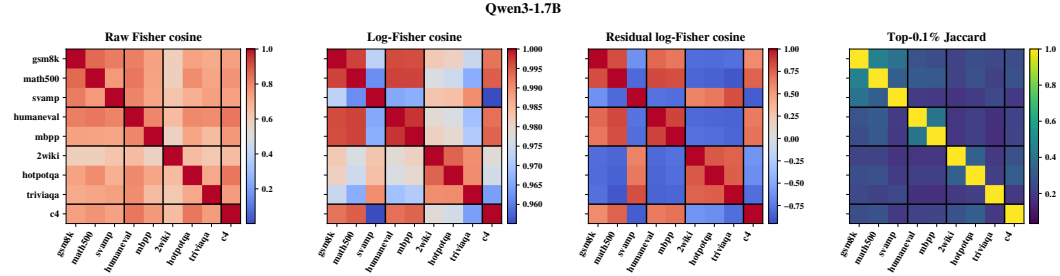


Figure 5: Capability regions in gradient space for Qwen3-1.7B: pairwise similarity of per-benchmark gradient signatures under four metrics (raw Fisher cosine, log-Fisher cosine, shared-component-removed log-Fisher cosine, top-0.1% coordinate Jaccard). Black lines separate the math, code, QA, and control blocks.

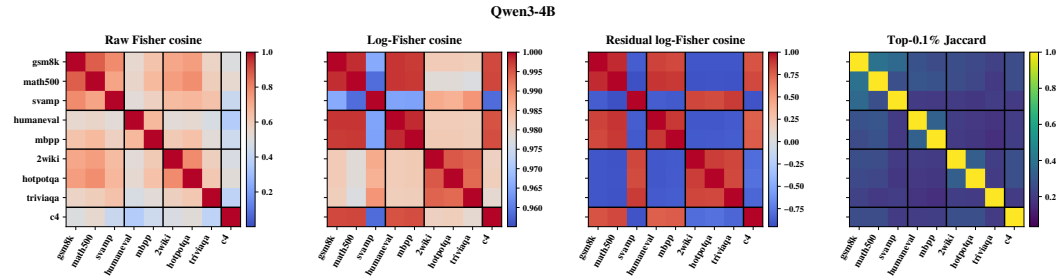


Figure 6: Capability regions in gradient space for Qwen3-4B: pairwise similarity of per-benchmark gradient signatures under four metrics (raw Fisher cosine, log-Fisher cosine, shared-component-removed log-Fisher cosine, top-0.1% coordinate Jaccard). Black lines separate the math, code, QA, and control blocks.

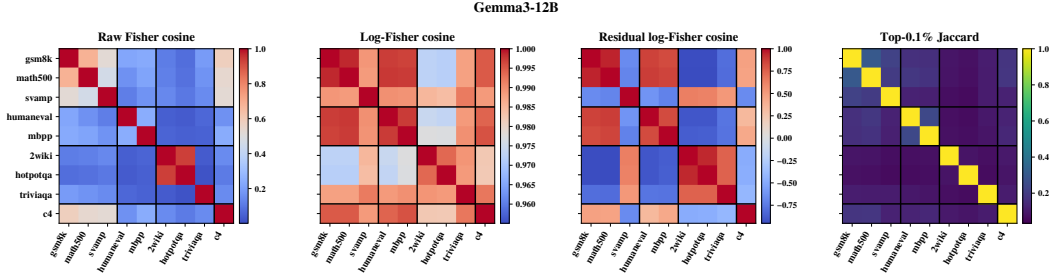


Figure 7: Capability regions in gradient space for Gemma3-12B: pairwise similarity of per-benchmark gradient signatures under four metrics (raw Fisher cosine, log-Fisher cosine, shared-component-removed log-Fisher cosine, top-0.1% coordinate Jaccard). Black lines separate the math, code, QA, and control blocks.

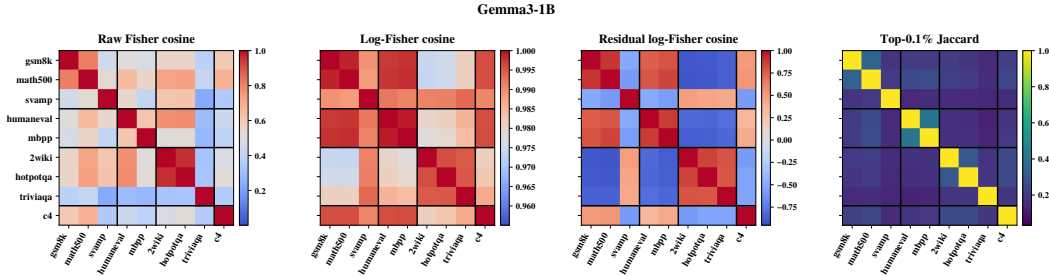


Figure 8: Capability regions in gradient space for Gemma3-1B: pairwise similarity of per-benchmark gradient signatures under four metrics (raw Fisher cosine, log-Fisher cosine, shared-component-removed log-Fisher cosine, top-0.1% coordinate Jaccard). Black lines separate the math, code, QA, and control blocks.

Selective ablation (preliminary evidence: single model; random, magnitude-matched, and global-Fisher controls are left to future work). A first targeted test: select the top 0.05% of weight coordinates by each capability’s residual Fisher signature (excluding coordinates shared with other capabilities) and zero them. The damage matrix is nearly diagonal (Gemma-3-1B; rows ablate, columns measure, nats/token):

Ablated region	math	code	QA
math	+4.63	+0.43	+0.05
code	+0.42	+7.14	+0.00
QA	+0.01	+0.00	+2.29

Zeroing 0.05% of weights, selected from gradients alone, selectively removes one capability while leaving the others nearly untouched (the QA ablation moves code by +0.0001). Pending multi-model replication and the control ablations above, we describe the regions as causally implicated and partially separable; we do not call them proven causal carriers; if confirmed, ablation itself is a form of targeted scaling-down.

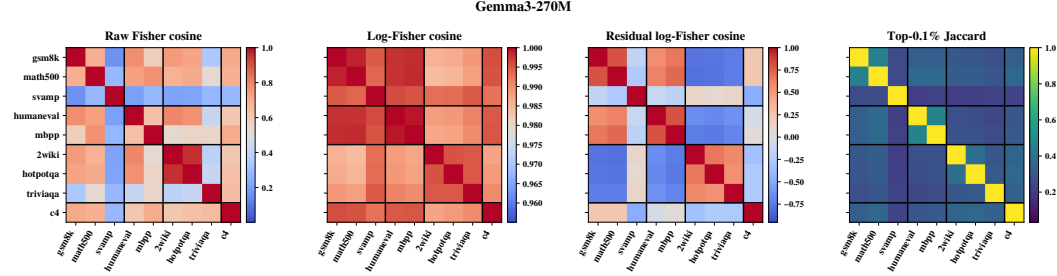


Figure 9: Capability regions in gradient space for Gemma3-270M: pairwise similarity of per-benchmark gradient signatures under four metrics (raw Fisher cosine, log-Fisher cosine, shared-component-removed log-Fisher cosine, top-0.1% coordinate Jaccard). Black lines separate the math, code, QA, and control blocks.

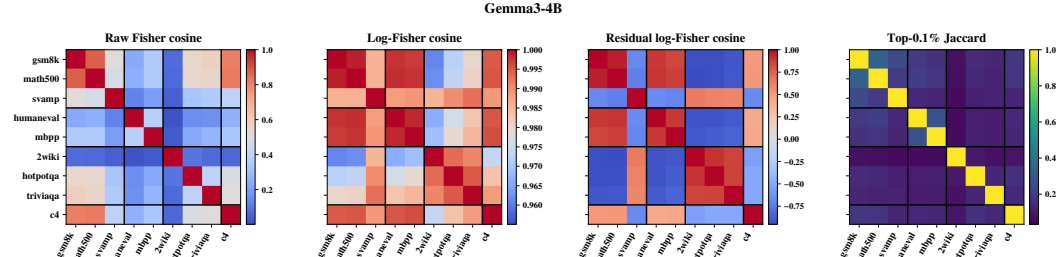


Figure 10: Capability regions in gradient space for Gemma3-4B: pairwise similarity of per-benchmark gradient signatures under four metrics (raw Fisher cosine, log-Fisher cosine, shared-component-removed log-Fisher cosine, top-0.1% coordinate Jaccard). Black lines separate the math, code, QA, and control blocks.

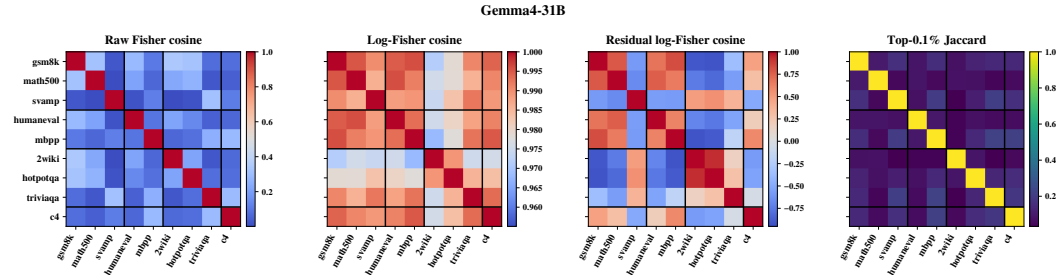


Figure 11: Capability regions in gradient space for Gemma4-31B: pairwise similarity of per-benchmark gradient signatures under four metrics (raw Fisher cosine, log-Fisher cosine, shared-component-removed log-Fisher cosine, top-0.1% coordinate Jaccard). Black lines separate the math, code, QA, and control blocks.

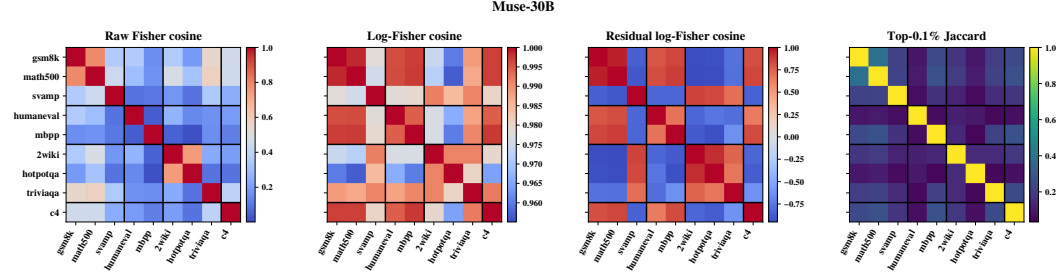


Figure 12: Capability regions in gradient space for Muse-30B: pairwise similarity of per-benchmark gradient signatures under four metrics (raw Fisher cosine, log-Fisher cosine, shared-component-removed log-Fisher cosine, top-0.1% coordinate Jaccard). Black lines separate the math, code, QA, and control blocks.

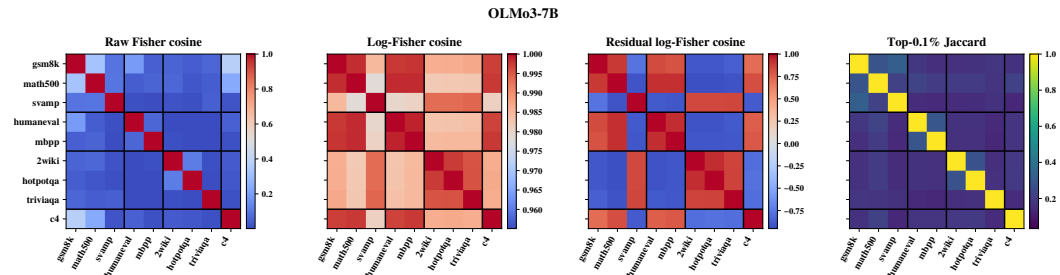


Figure 13: Capability regions in gradient space for OLMo3-7B: pairwise similarity of per-benchmark gradient signatures under four metrics (raw Fisher cosine, log-Fisher cosine, shared-component-removed log-Fisher cosine, top-0.1% coordinate Jaccard). Black lines separate the math, code, QA, and control blocks.

B.2 The pruning law

The mechanism form below fits the entire pre-cliff grid. With the measured alignment term subtracted, the curvature residual follows a power law in the deleted capability mass $m_c(d)$:

$$\Delta L_c(d) = \underbrace{\bar{g}_c^\top \delta w_d}_{\text{measured}} + B \left[\underbrace{m_c(d)}_{\text{measured}} \right]^\gamma, \quad (3)$$

with in-sample $R^2 = 0.82\text{--}0.996$ over 24 capability $\times$ model fits. The held-out battery (1000-resample cell bootstrap) is where the form is tested: in leave-one-family-out transfer the shared- γ mechanism form attains MAE 2.17 nats versus 10.3 for per-capability raw sparsity and 8.2 for the direct-damage baseline (both CIs exclude zero), so the deleted-mass coordinate is what transfers across families; it also beats the direct-damage baseline in fit-shallow/predict-deep (3.11 vs. 4.24) and raw sparsity in leave-largest-model-out (1.41 vs. 2.15), while its edge over raw sparsity within-model does not reach significance (without the first-order term, QA is unfittable, $R^2 \leq 0.67$; with it, QA joins the law). Two structural results: (i) in the deleted-mass coordinate the exponent concentrates at $\gamma \in [1.2, 2.2]$ across all families, versus 3.2–8.7 in the raw $(1-d)$ coordinate, which is what makes the form portable; and (ii) within a model, γ is shared across capabilities (e.g. Qwen3-0.6B: 1.41/1.56/1.43; Gemma3-4B: 1.66/1.71): capability differences appear to be carried by the two measured quantities, and the fitted structure is two parameters per model. This mechanism form uses gradient information of the dense model and is therefore not a K0 predictor; it is reported as an upper comparator for the reduced forms of the main text, not as a bound.

B.3 The signed first-order term predicts, in both directions

The alignment term $\bar{g}_c^\top \delta w$ is the family coefficient that carries sign. Its sign predicts the presence or absence of the capability-selective improvement in each case measured: the QA term of Qwen3 is negative and QA improves under pruning (predicted -0.83 vs. measured -0.79 at 4B, $d = 0.7$); the QA term of Gemma is positive and no improvement exists at any size; the term for OLMo-3-7B is positive (mild degradation, prediction $+0.015$ = measurement at $d = 0.8$, code) while the term for OLMo-3-32B is negative, and deep-pruning QA improvement duly appears. At the level of individual model $\times$ capability $\times$ density cells in the mild regime, sign prediction accuracy is 67.6% (Wilson 95% CI [58.3, 75.7], $n=108$), above chance, with the family-level pattern (Qwen and OLMo-32B QA improvements, Gemma none, an OLMo size flip) called in advance. Beyond the mild regime, first- plus second-order predictions fail by orders of magnitude at the cliff in each family: the collapse is non-perturbative in loss space, confirming the two-regime structure of equation 3.

B.4 The cliff is a geometric threshold crossing

Measuring the capability geometry of the pruned models accounts for the location of the cliff on the three models tested. The capability-region mass retention (the fraction of the capability-exclusive top coordinates of the dense model still prominent after pruning) declines smoothly from the first densities (no cliff in geometry) while loss is still flat; the behavioral cliff occurs where retention crosses a critical value. The critical value is family-specific (Gemma: 0.035–0.10 across 1B/4B and all capabilities; Qwen: 0.13–0.23) but consistent within a family across sizes and capabilities: the cliff boundary $d_c^*(N_0, f)$ reduces to one critical-retention constant per family, turning the least elegant fitted parameter of the two-regime law into a measurable, smoothly evolving order parameter. (preliminary evidence: three models, two families; a multi-model extension is left to future work)

B.5 Recovery arm: redeemable damage and the fitted plateau

Recovery training on general text fits the recovery law cleanly (Gemma3-270M pruned to $d = 0.6$; $R^2 \approx 1.0$): $\beta'_{\text{math}} = 1.23$, $\beta'_{\text{code}} = 1.02$, with fitted plateaus under the fixed LoRA recipe and the largest measured budget of 16M tokens, $L_c^{\text{plateau}} - L_c^{\text{dense}}$, of 1.21 (math) and 1.93 (code) nats; about 80% of this pruning damage is redeemed within 8M tokens and the remainder persists up to the largest budget measured under this recipe, while QA recovers

completely. The plateau is a fitted quantity at 16M tokens, not an established asymptote, and the saturating form does not describe the non-monotone aligned-data trajectory below. (preliminary evidence: one model, one density, one recipe; with three-point ladders the largest-budget extrapolation of the saturating form beats a carry-forward baseline in only 4 of 9 fits, so the ladder is too sparse to claim extrapolation, and a denser multi-model grid is left to future work) Two mechanism results: recovery on aligned data (teacher traces) heals substantially faster at small budgets (math 2.76 vs. 3.47 at 0.5M tokens), as the gradient-overlap prediction requires, but over-training on narrow aligned data re-damages the model (QA 4.95 $\rightarrow$ 9.75 from 0.5M to 8M tokens), while general-text recovery is monotone. Aligned recovery is fast at small budgets and harmful when over-trained on narrow data; general recovery is slow and monotone on this model. Quantization damage past the bit cliff is markedly less redeemable under the same recipe (residuals $\approx$ 2.6–2.8 after int3) than pruning damage at matched severity.

B.6 Scale and family structure

Within-family robustness is non-monotonic in two families independently. Gemma-3 robustness peaks at 12B (27B collapses far earlier, +4.0 vs. +0.68 at $d = 0.7$), mirroring the prior ability-space observation of a Qwen3 14B peak. A mid-scale robustness peak is a reproducible pattern of scaling-down.

A candidate predictor, honestly retired. The top-concentration statistic κ_c tracked robustness within the Qwen3 ladder and was pre-registered as a predictor; the full grid retires it (cross-family inversion: OLMo, the most robust family, has the lowest concentration; within-Gemma top-end violation).

Matched-size panel ($\sim$ 30B). At matched scale the family coefficient dominates: ΔL_{math} at $d = 0.7$ spans +0.02 (OLMo-3-32B), +0.17 (Muse-30B), +0.68 (Gemma4-31B), +4.04 (Gemma3-27B). OLMo-3-32B survives 70% pruning with $\leq$ +0.63 nats on each capability. Family identity is a first-order determinant of prunability on this panel; which training-data or recipe properties carry it was not measured.

B.7 New-source prospectives, second round

The second frozen round (Table 9) added seven pruning densities (0.9, 0.8, 0.75, 0.7, 0.65 and 0.6 as interpolation, 0.55 as extrapolation) and five bit-widths (8, 6, 5, 4, 3) on Pythia-1B at steps 32k and 112k. Two fitting protocols were frozen at the same time. Protocol A fits each candidate on the full nine-state development panel. Protocol B fits on development states at step 64k or below, so that its 112k rows test a training-stage extrapolation on top of the size change. The 5-bit prediction of the class-indicator candidate is a fixed rule, the average of its committed 4- and 6-bit predictions, chosen so that no 5-bit measurement enters any fit. Candidate names follow §4.2: power is the shared power form, A2 the per-density source regression with a fixed interpolation rule, A1 its $\gamma = 1$ variant, cont a continuous two-term polynomial in $1 - d$, and the three source-free baselines are strength-only, the median development curve, and zero change. For quantization, full uses $\{N_0, L_0, D_0\}$, no-D0 drops the training-stage term, and mean and median are per-bit constants from the development panel.

Per capability, at 112k under protocol A the source regression reaches MAE 0.111 (math), 0.091 (code) and 0.135 (QA); the strongest source-free alternative is strength-only for math (0.117) and the median curve for code (0.107) and QA (0.261). At 32k the QA error of the source regression rises to 0.339 while the median curve stays at 0.193, which accounts for most of the aggregate gap. Under protocol B at 112k the power and A2 forms are ahead of strength-only at $d = 0.55$ (0.96 and 1.00 against 1.33), but with errors near one nat the ordering carries little practical value. The pair cost 22 A100 minutes.

The 6.9B pair, a size extrapolation of about five times, was measured under the same freeze (36 A100 minutes, reference model on CPU). Its dense anchors are math 1.27, code 1.30 and QA 5.11 at 32k, and 1.11, 1.21 and 5.09 at 112k. Per capability under protocol A at 112k,

the source regression reaches 0.23 (math), 0.19 (code) and 0.96 (QA) against 0.05, 0.07 and 0.15 for the median curve. Under protocol B the source forms lead on code only (A2 0.058 and 0.083 at the two stages against 0.208 and 0.074 for the median curve) and trail on QA (0.32 and 0.20 against 0.08 and 0.15). At $d = 0.55$ the source-free curves win under both protocols, with source-conditioned errors between 0.55 and 2.2 nats. For quantization under protocol A the class-indicator candidate is worst or near worst above 3 bits (0.25 and 0.27 against 0.03 and 0.04 for the per-bit median) and overshoots by about four nats at 3 bits. Protocol B compresses the differences above 3 bits to the 0.02 to 0.05 range and leaves the candidate ahead at 3 bits only for the late stage (1.29 against 3.8 to 4.1).

B.8 Re-analysis of the prior ability-space grid

A prior internal study fit compression laws directly on Rasch abilities from ~ 100 -item panels over a 774-cell Qwen3 grid and reported that no derived law had held-out predictive power. Our re-analysis attributes much of this to the measurement layer: (1) the $r=1$ falsification anchor was driven almost entirely by the answer-only recipe (stratified, the anchor holds and held-out error drops from $9.0\times$ to $1.8\times$ the noise floor); (2) same-benchmark evaluation inflated QA gains by roughly one third; (3) discrimination heterogeneity voids interval-scale ability while rankings survive ($\rho \geq 0.98$); (4) item-level damage is smooth pre-cliff, ruling out discrete skill deletion at the item level. These findings fixed the protocol of §3.

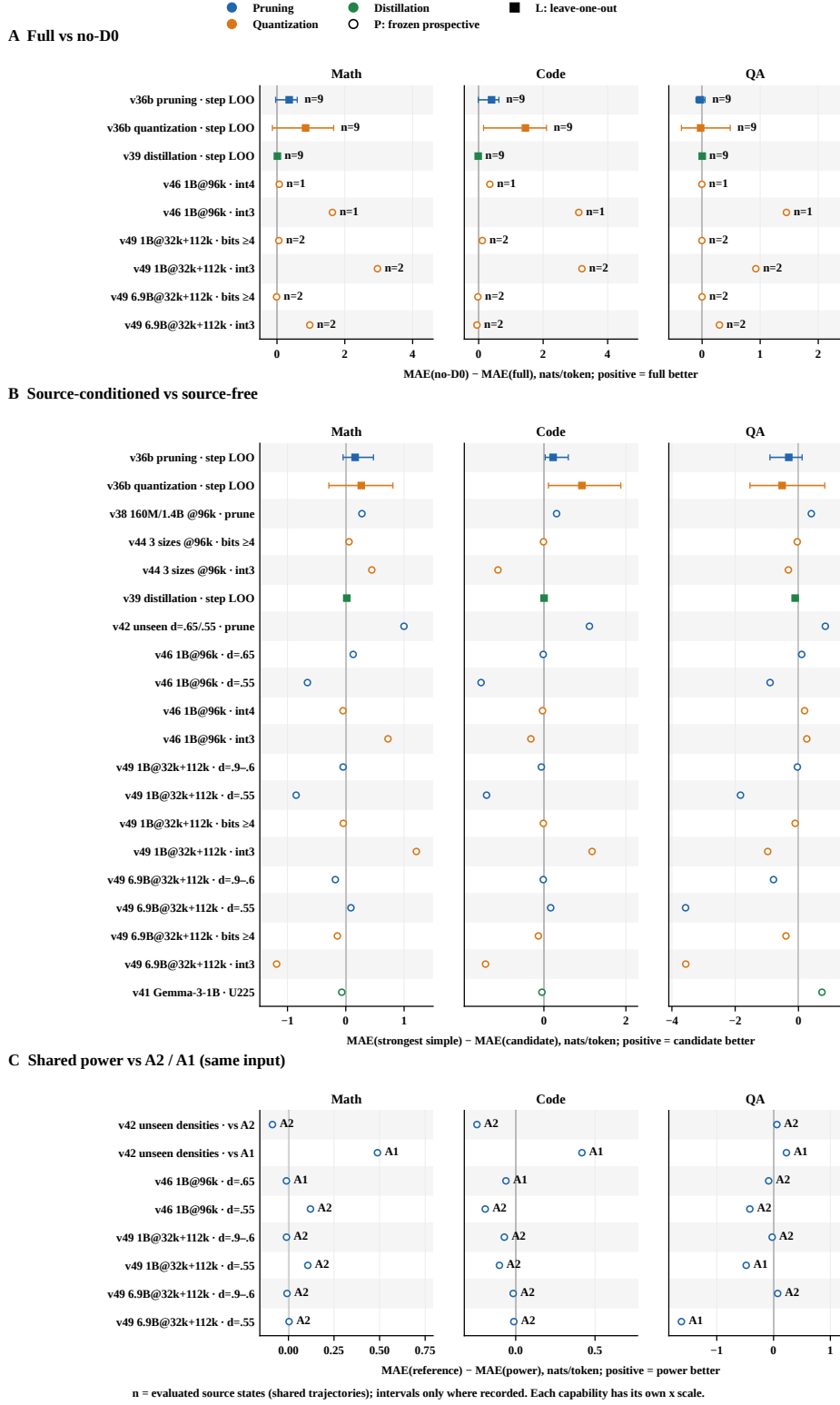


Figure 14: Paired improvements, zero line marked, positive means the first-named model is better. A: full model against its no- D_0 variant, per arm and capability; intervals only where a cluster bootstrap was computed (leave-one-stage-out panels), n = evaluated source states. B: source-conditioned candidate against the strongest source-free baseline for all twenty registered tests (hollow = frozen prospective, filled = leave-one-out). C: shared power form against the same-input alternatives A1 and A2 on the pruning configuration tests. Generated by plot_fig2_gains.py from [27](#) v52 groups and the v36b/v39/v42/v46/v49 result files.

Test; unseen axis; [origin]	Line	math	code	QA
pruning: source transfer at fixed d : 3×3 Pythia panel, leave-one-step-out held-out pretraining stage (in-range) [L · F · R · A]	cand.: config-indicator OLS	0.253 (+0.16)	0.292 (+0.22)*	0.806 (-0.31)
	same-input: no- D_0	0.613	0.693	0.774
	simple: per-config	0.413	0.516	0.500
	median; zero change			
quantization: source transfer at fixed b : 3×3 Pythia panel, leave-one-step-out held-out pretraining stage (in-range) [L · F · R · A]	cand.: config-indicator OLS	1.144 (+0.27)	0.626 (+0.93)*	1.989 (-0.52)
	same-input: no- D_0	1.989	2.077	1.964
	simple: per-config	1.410	1.553	1.473
	median			
pruning: new stage 96k on 160M and 1.4B, all measured d stage 96k (interpolation between 64k and 143k) [P · F · R · A]	cand.: config-indicator OLS	0.090 (+0.25)	0.128 (+0.05)	0.426 (+0.01)
	same-input: no- D_0	0.343	0.179	0.438
	simple: per-config	0.367	0.438	0.832
	median			
quantization: new stage 96k on 160M, 410M, 1.4B; $b \geq 4$ (near-zero regime) stage 96k (interpolation) [P · - · R · A]	cand.: config-indicator OLS	0.029 (+0.06)	0.104 (-0.01)	0.173 (-0.04)
	same-input: none available	-	-	-
	simple: per-bit	0.084	0.094	0.134
	median			
quantization: new stage 96k on 160M, 410M, 1.4B; int3 (collapse regime) stage 96k (interpolation) [P · - · R · A]	cand.: config-indicator OLS	1.488 (+0.45)	1.963 (-1.12)	2.046 (-0.32)
	same-input: none available	-	-	-
	simple: per-bit	1.934	0.843	1.727
	median			
distillation: δ_c source transfer: 3×3 LoRA panel, leave-one-step-out held-out pretraining stage (in-range) [L · F · R · A]	cand.: linear { N_0, L_0, D_0 }	0.014 (+0.01)	0.057 (-0.01)	0.263 (-0.10)
	same-input: no- D_0	0.026	0.044	0.265
	simple: constant (mean); constant (median)	0.031	0.061	0.159

Table 6: Source axis: does the pre-compression source state predict the response at a fixed intervention setting? MAE in nats/token per capability; bold = smallest in the group; the number in parentheses is the improvement of the candidate over the strongest listed baseline (positive = candidate better); * = bootstrap 95% interval of that improvement excludes zero (computed only for the leave-one-out panels). Origin code: prediction origin (P frozen prospective, L leave-one-out) · same-input baseline origin · simple baseline origin (F pre-specified with the candidate, R computed after the test, - none) · selection (A: all pre-specified forms reported). Where several same-input or simple baselines exist, the strongest per capability is shown and named; that choice is retrospective. Config-indicator OLS uses $\{N_0, L_0, D_0\}$ with one coefficient set per configuration. Generated by v52_prediction_tables.py.

Test; unseen axis; [origin]	Line	math	code	QA
pruning: unseen d 0.65 and 0.55, dev sources + 410M@96k densities 0.65 (interp.) and 0.55 (extrap.) pooled [P · R · F · A]	cand.: shared power form	0.371 (-0.09)	0.524 (-0.24)	0.759 (+0.06)
	same-input: A2 (per- d regr.)	0.283	0.279	0.818
	simple: strength-only	1.369	1.635	1.608
pruning: new source 1B@96k, $d = 0.65$ (interp.) in-range size; stage 96k [P · F · F · A]	cand.: shared power form	0.020 (-0.01)	0.092 (-0.06)	0.466 (-0.09)
	same-input: A1 ($\gamma=1$); A2 (per- d regr.)	0.009	0.031	0.380
	simple: median curve	0.147	0.076	0.569
pruning: new source 1B@96k, $d = 0.55$ (extrap.) in-range size; stage 96k [P · F · F · A]	cand.: shared power form	0.930 (-0.66)	1.550 (-1.53)	1.426 (-0.90)
	same-input: A2 (per- d regr.)	1.051	1.358	1.009
	simple: strength-only	0.273	0.018	0.530
pruning: new source 1B@32k+112k, d 0.9–0.6 (interp.) in-range size; stages 32k, 112k [P · F · F · A]	cand.: shared power form	0.215 (-0.05)	0.191 (-0.07)	0.261 (-0.03)
	same-input: A2 (per- d regr.)	0.205	0.119	0.237
	simple: median curve; strength-only	0.169	0.128	0.227
pruning: new source 1B@32k+112k, $d = 0.55$ (extrap.) in-range size; stages 32k, 112k [P · F · F · A]	cand.: shared power form	1.790 (-0.85)	1.488 (-1.40)	2.225 (-1.83)
	same-input: A1 ($\gamma=1$); A2 (per- d regr.)	1.896	1.386	1.744
	simple: strength-only	0.939	0.089	0.393
pruning: new source 6.9B@32k+112k, d 0.9–0.6 (interp.) size $5\times$ beyond dev range; stages 32k, 112k [P · F · F · A]	cand.: shared power form	0.299 (-0.18)	0.160 (-0.02)	0.909 (-0.79)
	same-input: A2 (per- d regr.)	0.292	0.144	0.982
	simple: median curve	0.120	0.144	0.120
pruning: new source 6.9B@32k+112k, $d = 0.55$ (extrap.) size $5\times$ beyond dev range; stages 32k, 112k [P · F · F · A]	cand.: shared power form	0.615 (+0.00)	0.547 (-0.01)	4.472 (-3.57)
	same-input: A1 ($\gamma=1$); A2 (per- d regr.)	0.618	0.535	2.850
	simple: median curve; strength-only	0.703	0.712	0.904

Table 7: Configuration axis, pruning: does a relation fit at seen densities and sources predict unseen densities, sizes and stages? Protocol A (full development panel) for the two P1-v2 pairs and the single-stage 1B@96k prospective; protocol B in Table 9. Format, bold, improvement in parentheses, and origin code as in Table 6.

Test; unseen axis; [origin]	Line	math	code	QA
quantization: new source 1B@96k, int4 in-range size; stage 96k; seen bit $[P \cdot F \cdot F \cdot A]$	cand.: config-indicator OLS	0.116 (-0.05)	0.046 (-0.03)	0.059 (-0.00)
	same-input: no- D_0	0.184	0.394	0.054
	simple: per-bit median	0.068	0.012	0.253
quantization: new source 1B@96k, int3 in-range size; stage 96k; seen bit $[P \cdot F \cdot F \cdot A]$	cand.: config-indicator OLS	0.690 (+0.72)	0.318 (-0.32)	0.182 (+0.26)
	same-input: no- D_0	2.324	3.423	1.635
	simple: per-bit median	1.414	0.001	0.446
quantization: new source 1B@32k+112k, $b \geq 4$ (incl. 5-bit rule) in-range size; stages 32k, 112k $[P \cdot F \cdot F \cdot A]$	cand.: config-indicator OLS	0.064 (-0.04)	0.041 (-0.01)	0.157 (-0.10)
	same-input: no- D_0	0.118	0.156	0.153
	simple: per-bit median	0.020	0.028	0.052
quantization: new source 1B@32k+112k, int3 in-range size; stages 32k, 112k $[P \cdot F \cdot F \cdot A]$	cand.: config-indicator OLS	0.256 (+1.21)	0.362 (+1.18)	1.972 (-0.97)
	same-input: no- D_0	3.218	3.565	2.897
	simple: per-bit mean; per-bit median	1.466	1.537	1.000
quantization: new source 6.9B@32k+112k, $b \geq 4$ (incl. 5-bit rule) size $5\times$ beyond dev range; stages 32k, 112k $[P \cdot F \cdot F \cdot A]$	cand.: config-indicator OLS	0.181 (-0.14)	0.177 (-0.13)	0.428 (-0.39)
	same-input: no- D_0	0.169	0.154	0.426
	simple: per-bit median; zero change	0.036	0.043	0.034
quantization: new source 6.9B@32k+112k, int3 size $5\times$ beyond dev range; stages 32k, 112k $[P \cdot F \cdot F \cdot A]$	cand.: config-indicator OLS	3.233 (-1.19)	3.229 (-1.42)	5.306 (-3.56)
	same-input: no- D_0	4.199	3.176	5.605
	simple: per-bit mean	2.046	1.805	1.744
distillation: unseen pool U225 from endpoint pools U75/U600 (Gemma-3-1B) pool (reuse count in-range; sampling protocol differs) $[P \cdot F \cdot F \cdot A]$	cand.: E -only (reuse count)	0.191 (-0.07)	0.117 (-0.05)	0.706 (+0.00)
	same-input: $T+E$ joint	0.215	0.145	0.710
	simple: constant	0.122	0.069	1.449

Table 8: Configuration axis, quantization and distillation: unseen sizes and stages at seen bit-widths (protocol A; the 5-bit prediction is a fixed interpolation rule), and an unseen distillation pool. Format, bold, improvement in parentheses, and origin code as in Table 6.

Source	Prot. Regime		power	A2	A1	cont	str.-only	median	zero	best	
1b@32000	A	prune interp	d0.9-0.6	0.254	0.261	0.393	0.275	0.363	0.204	0.412	median-curve
1b@32000	A	prune extrap	d0.55	2.272	2.130	2.394	2.221	0.637	1.509	2.434	strength-only
1b@32000	B	prune interp	d0.9-0.6	0.249	0.218	0.321	0.239	0.236	0.224	0.412	A2
1b@32000	B	prune extrap	d0.55	1.772	1.777	2.151	1.863	1.531	1.645	2.434	strength-only
1b@112000	A	prune interp	d0.9-0.6	0.191	0.112	0.300	0.175	0.365	0.173	0.400	A2
1b@112000	A	prune extrap	d0.55	1.398	1.235	1.838	1.356	0.311	1.311	2.237	strength-only
1b@112000	B	prune interp	d0.9-0.6	0.223	0.177	0.355	0.235	0.219	0.195	0.400	A2
1b@112000	B	prune extrap	d0.55	0.957	0.995	1.685	1.127	1.333	1.447	2.237	power
6.9b@32000	A	prune interp	d0.9-0.6	0.476	0.482	0.608	0.501	0.483	0.165	0.160	zero
6.9b@32000	A	prune extrap	d0.55	1.635	1.423	0.939	1.457	1.493	0.275	0.889	median-curve
6.9b@32000	B	prune interp	d0.9-0.6	0.160	0.166	0.249	0.187	0.163	0.136	0.160	median-curve
6.9b@32000	B	prune extrap	d0.55	0.705	0.549	0.677	0.630	0.252	0.148	0.889	median-curve
6.9b@112000	A	prune interp	d0.9-0.6	0.436	0.463	0.610	0.479	0.376	0.092	0.294	median-curve
6.9b@112000	A	prune extrap	d0.55	2.121	1.939	2.224	2.061	0.191	1.272	2.198	strength-only
6.9b@112000	B	prune interp	d0.9-0.6	0.131	0.106	0.235	0.140	0.111	0.095	0.294	median-curve
6.9b@112000	B	prune extrap	d0.55	1.441	1.387	1.892	1.525	1.294	1.408	2.198	strength-only
Source	Prot. Regime		full	no-D0	mean	median	zero	best			
1b@32000	A	quant ≥ 4 -bit	0.103	0.174	0.197	0.031	0.046	per-bit-median			
1b@32000	A	quant int3	1.050	2.648	2.903	1.438	3.108	full			
1b@32000	A	quant int5 (rule)	0.179	0.272	0.254	0.044	0.017	zero			
1b@32000	B	quant ≥ 4 -bit	0.028	0.032	0.027	0.026	0.046	per-bit-median			
1b@32000	B	quant int3	0.606	1.004	0.627	0.803	3.108	full			
1b@32000	B	quant int5 (rule)	0.025	0.013	0.035	0.034	0.017	noD0			
1b@112000	A	quant ≥ 4 -bit	0.071	0.111	0.183	0.036	0.063	per-bit-median			
1b@112000	A	quant int3	0.676	3.805	0.802	1.231	5.777	full			
1b@112000	A	quant int5 (rule)	0.151	0.138	0.251	0.035	0.025	zero			
1b@112000	B	quant ≥ 4 -bit	0.038	0.042	0.038	0.039	0.063	per-bit-mean			
1b@112000	B	quant int3	1.120	1.981	3.106	3.472	5.777	full			
1b@112000	B	quant int5 (rule)	0.046	0.019	0.027	0.025	0.025	noD0			
6.9b@32000	A	quant ≥ 4 -bit	0.253	0.317	0.190	0.034	0.033	zero			
6.9b@32000	A	quant int3	3.973	5.103	3.300	1.835	2.711	per-bit-median			
6.9b@32000	A	quant int5 (rule)	0.373	0.457	0.237	0.067	0.029	zero			
6.9b@32000	B	quant ≥ 4 -bit	0.033	0.050	0.029	0.025	0.033	per-bit-median			
6.9b@32000	B	quant int3	0.489	1.792	0.174	0.407	2.711	per-bit-mean			
6.9b@32000	B	quant int5 (rule)	0.046	0.052	0.058	0.057	0.029	zero			
6.9b@112000	A	quant ≥ 4 -bit	0.271	0.183	0.155	0.042	0.061	per-bit-median			
6.9b@112000	A	quant int3	3.873	3.550	0.430	1.896	6.442	per-bit-mean			
6.9b@112000	A	quant int5 (rule)	0.402	0.247	0.253	0.033	0.027	zero			
6.9b@112000	B	quant ≥ 4 -bit	0.048	0.060	0.042	0.045	0.061	per-bit-mean			
6.9b@112000	B	quant int3	1.294	4.033	3.770	4.137	6.442	full			
6.9b@112000	B	quant int5 (rule)	0.043	0.025	0.025	0.023	0.027	per-bit-median			

Table 9: P1-v2 new-source frozen perspectives (MAE over three capabilities, nats/token). Protocol A fits on the full nine-state development panel; protocol B on step $\leq 64k$ only, so its 112k rows are training-stage extrapolations. 1B sources are in-range sizes and 6.9B is a $\sim 5\times$ size extrapolation. Predictions were committed before measurement; per-capability and per-point values are in the results ledger.

Candidate	P	test MAE			signed bias (pred. – obs.)		
		math	code	QA	math	code	
power (selected)	5	0.243	0.236	0.678	+0.216	+0.236	
A2	20	0.230	0.222	0.682	+0.222	+0.212	
A1	4	0.363	0.403	0.738	+0.210	+0.206	
cont	8	0.313	0.305	0.746	+0.116	+0.107	
strength only	2	0.450	0.488	0.579	+0.443	+0.486	
median curve	0	0.277	0.214	0.221	+0.277	+0.214	
zero	0	0.522	0.530	0.185	-0.522	-0.530	
gain of the selected form vs A2 (same input): math -0.013, code -0.014, qa +0.004; vs strongest source-free: math +0.034, code +0.034, qa +0.004							

Table 10: Round-3 pruning deliverable. Inputs $\phi = [1, \log \tilde{N}_0, \tilde{L}_{0,c}, \log \tilde{D}_0]$ (standardized on the development set); development set = 17 Pythia states (160M–6.9B, steps 16k–143k, densities ≥ 0.55); the selected form was chosen by a pre-committed leave-one-source-out rule (tie within 0.02 nats resolved toward the continuous, zero-at- $d=1$ form with fewer parameters). Test = three new checkpoints (410M@48k, 1.4B@112k, 6.9B@80k) at $d \in \{0.85, 0.675, 0.575\}$, predictions committed before measurement; MAE and mean signed error (prediction minus observation) over nine cells per capability. Coefficients in Appendix Table 11. Generated by v57_round3_tables.py.

capability	γ_c	$\beta_{c,0}$	$\beta_{c,\log N_0}$	β_{c,L_0}	$\beta_{c,\log D_0}$
math	3.75	+3.275	+0.319	+3.850	+0.250
code	3.05	+4.499	+0.273	+5.357	+0.347
qa	3.05	-7.409	-0.481	+5.486	+0.106

Table 11: Selected pruning form $\widehat{\Delta L}_c = (\beta_c \cdot \phi)((1-d)/0.3)^{\gamma_c}$ with $\phi = [1, z(\log N_0), z(L_{0,c}), z(\log D_0)]$; standardization centers 20.248, 2.661, 25.614 and scales 1.407, 1.691, 0.739 for $(\log N_0, L_{0,c}, \log D_0)$ in natural units (N_0 = transformer weight-matrix parameters, D_0 = tokens). Register: v53-prune-dev/register.json.

Candidate	P	bit test (12)			granularity test (18)			joint, 1B@96k (3)		
		math	code	QA	math	code	QA	math	code	QA
low-order 2D	20	0.193	0.214	0.436	0.219	0.319	0.986	0.345	0.097	
separable	6	0.767	0.798	1.427	0.581	0.643	1.612	0.586	0.380	
same input interpolation	16	1.296	1.381	1.942	0.606	0.711	1.497	0.538	0.241	
median	0	0.553	0.726	0.537	1.123	1.222	1.220	0.152	0.270	
mean	0	1.416	1.556	1.495	1.857	1.906	2.019	1.491	1.558	
zero	0	0.450	0.532	0.480	1.239	1.346	1.239	0.300	0.339	
signed bias of the 2D form (math/code/QA): bit +0.108/+0.060/+0.077; granularity +0.162/+0.137/+0.129; joint +0.345/+										

Table 12: Round-3 grouped-quantization deliverable (MAE, nats). Same inputs ϕ as Table 10; development = six Pythia states (160M/410M/1.4B at steps 16k and 143k) $\times b \in \{3, 5\} \times g \in \{64, 256\}$ under symmetric round-to-nearest with contiguous input groups of g weights; all predictions committed before the tests. Bit test: unseen $b=4$ at $g \in \{64, 256\}$ on the dev states; granularity test: unseen $g=128$ at $b \in \{3, 4, 5\}$ on the dev states; joint test: new state 1B@96k at $g=128$. All candidates and baselines are reported on every test (pre-committed rule; no selection by outcome). Generated by v57_round3_tables.py.